CHAPTER 2:

COMPUTER SYSTEMS

BSAIS 3A

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COMPUTER SYSTEM

a computer system is a complex machine made up of a ***combination of hardware and software components***. These components work together to perform specific tasks and store data.

HARDWARE

physical pieces of a computer system—such as a central processing unit (CPU) , a printer, and a disk drive—that can be touched.

SOFTWARE

which is the set of programs that controls the operations of the computer system

TWO DISTINCT TYPES OF COMPUTERS

DIGITAL

operate directly on numbers, or digits, just as humans do.

ANALOG

manipulate some analogous physical quantity, such as voltage or shaft rotation speed, which represents the numbers involved in the computation.

BASIC COMPONENTS OF COMPUTER SYSTEMS

- Input
- Output
- Computer Memory
- Arithmetic/Logical Unit
- Computer Files
- Control Unit

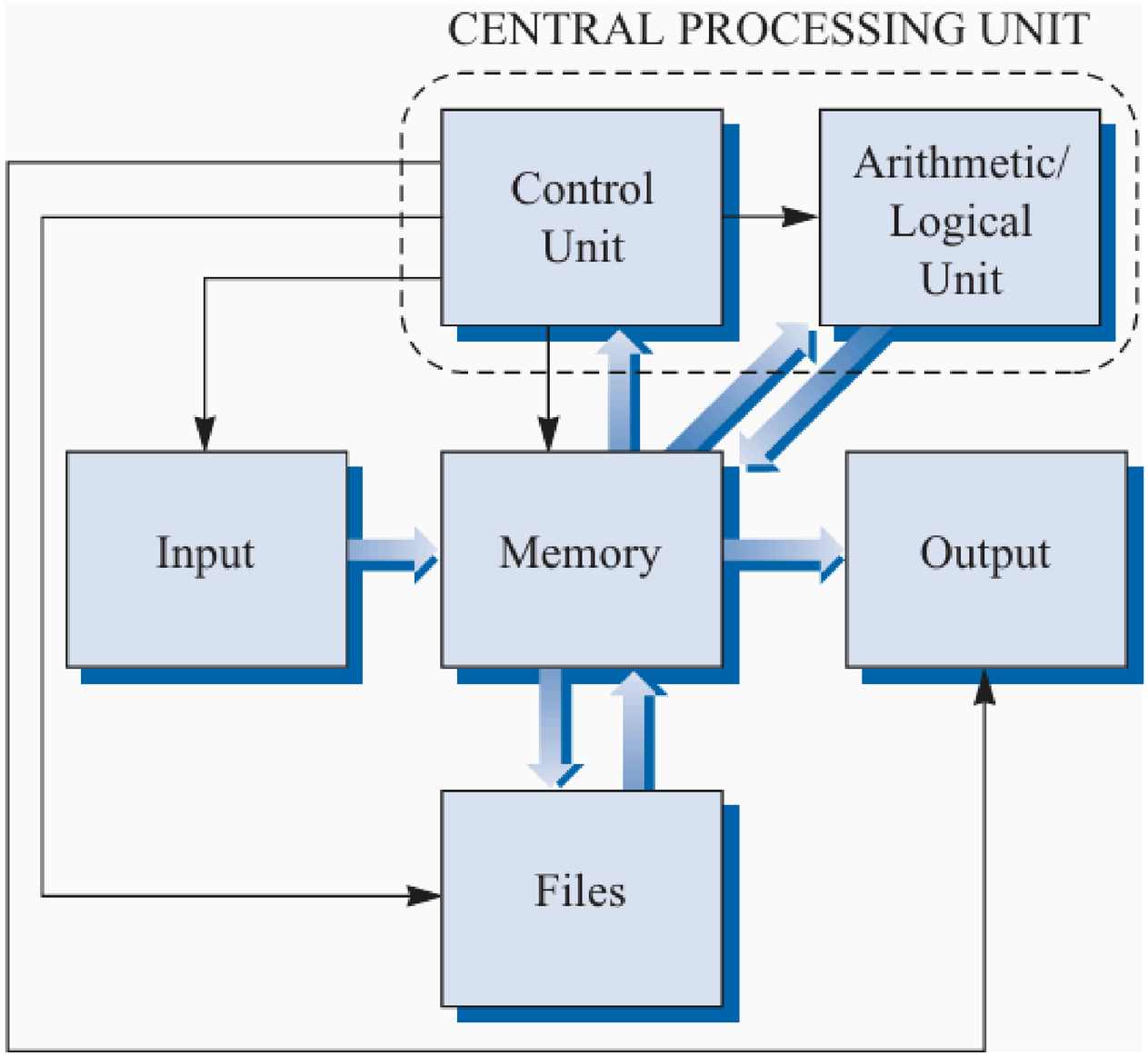
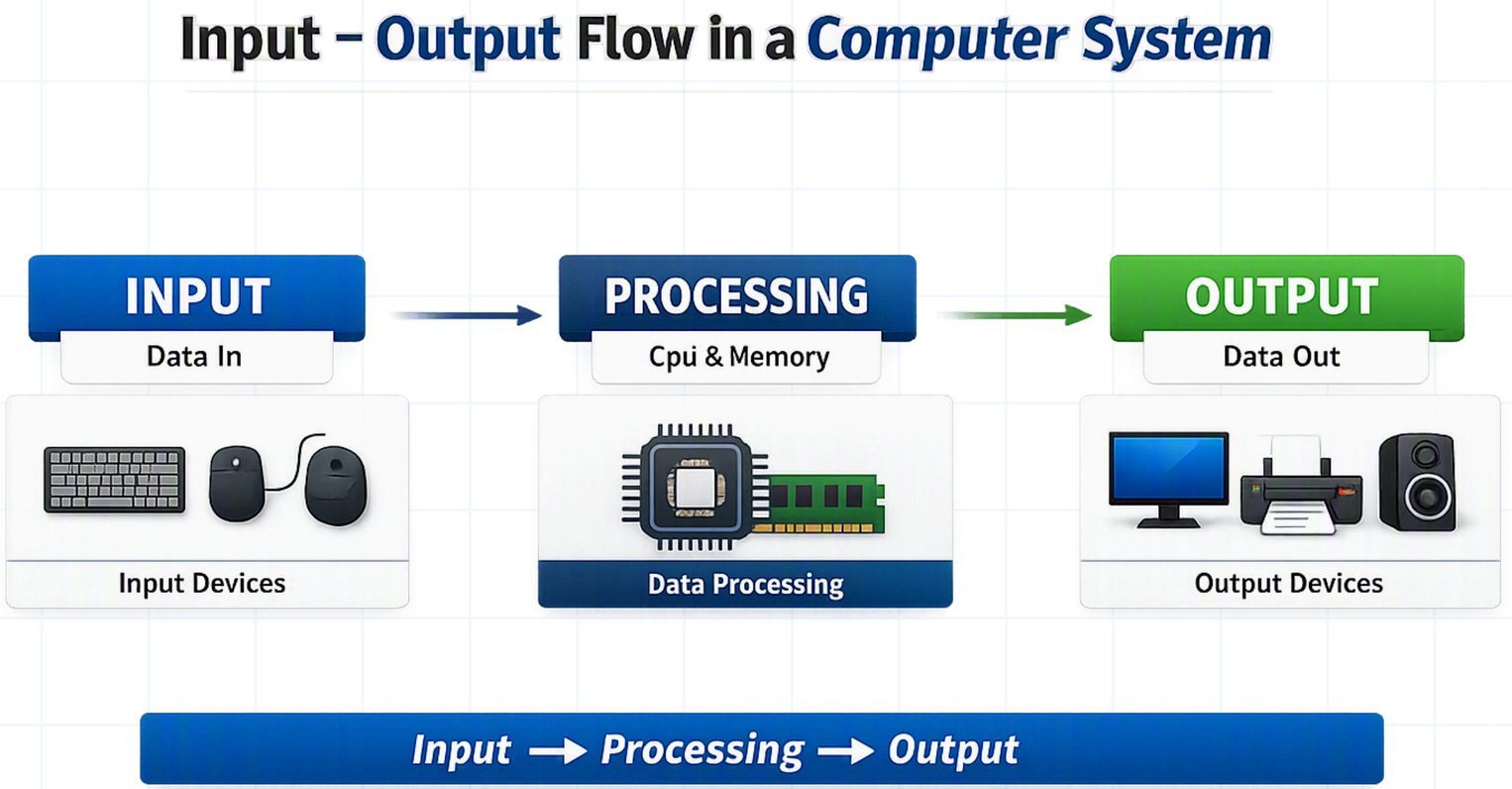


FIGURE 2.1 The Logical Structure of Digital Computers

INPUT/OUTPUT

- how a computer communicates with users
- input is the data enetered into a computer, while output is the results produce.



INPUT/OUTPUT

Input devices:

- Keyboard and mouse
- Terminals
- Voice input
- MICR (Magnetic Ink Character Recognition)
- OCR (Optical Character Recognition)
- Imaging scanners
- Bar code scanners

Output Devices:

- Monitors (video display units)
- Printers –
- CD/DVD drives
- Computer Output Microfilm (COM)
- Voice response units

COMPUTER MEMORY

- divided into cells, and a fixed amount of data (number, letter, words, or symbol) can be stored in each cell.
- each memory cell has an identifying number, called an *address*, which never changes.

COMPUTER MEMORY

Memory

WORD **BYTE**

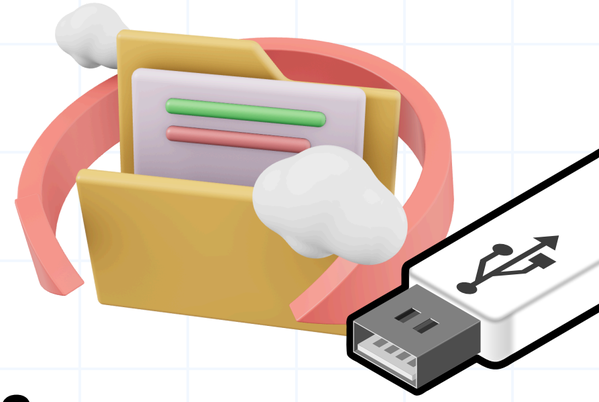
0	1	2	3	4	5	6	7
MAY	2012	700.00	4	OSU	17	321.16	3
8	9	10	11	12	13	14	15
C	OMPU	TER	32	0	MARY	71.3	L
16	17	18	19	20	21	22	23
27	18	103.0	7	JOHN	41	100.00	0
24	25	26	27	28	29	30	31
0	0	0	37	B	0	62	1

FIGURE 2.2 Diagram of Computer Memory

Post Office Boxes

COMPUTER FILES

- sometimes called *secondary memory or secondary storage devices.*
- it includes magnetic tape drives, hard (or fixed) disk drives, removable disk drives, flash drives, and CD or DVD (optical) drives.



TWO BASIC WAYS TO ORGANIZE COMPUTER FILES:

SEQUENTIAL ACCESS

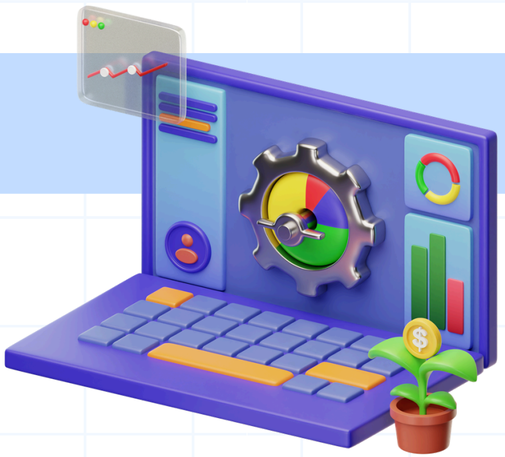
files are stored in sequence according to the file's control key.

ex. payroll file

DIRECT ACCESS FILES

Stored on a *direct access storage device (DASD)*, is a file from which it is possible for the computer to obtain a record immediately, without regard to where the record is located in the file.

ARITHMETIC/LOGICAL UNIT



- performs calculations on data received from memory or registers.

FUNCTIONS OF ALU;

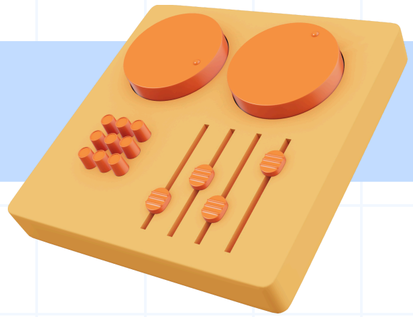
Arethmetic Operations

- *addition*
- *subtraction*
- *multiplication*
- *division*

Logical Operation

- *AND*
- *OR*
- *NOT*
- < > =

CONTROL UNIT



- Part of the Central Processing Unit (CPU)
- Controls the operation of all major computer components.
- Retrieves (fetches) instructions from memory.
- Interprets (decodes) each instruction.
- Directs the Arithmetic/Logic Unit (ALU), memory, and input/output devices to perform the required tasks.

THE STORED-PROGRAM CONCEPT

- the principle that a computer stores both programs (instructions) and data in its memory. The computer's control unit reads and executes these instructions one at a time at electronic speed.

TYPES OF COMPUTER SYSTEM

1. Microcomputers
2. Midrange systems
3. Mainframes
4. Supercomputers

MICROCOMPUTERS

FIGURE 2.5 HP EliteBook 2740p Tablet PC (Courtesy of Hewlett-Packard)

- often called micros or personal computers or just PCs
- single keyboard & display
- netbooks ($300–$500)
- IBM introduced the original Personal Computer
- HP, Dell, Acer, Lenovo, Toshiba, Fujitsu, and Sony.

(COST: $200–$4,000)

COMMON TYPES OF PC:

- Desktop PCs
- Laptops/Notebooks
- Handheld/Palmtop PCs
- Smartphones
- Tablet PCs
- Netbooks

MIDRANGE SYSTEM

- a computer that is more powerful than a microcomputer (PC) but less powerful than a mainframe
- workstations and minicomputers.
- development of RISC (Reduced Instruction Set Computing) processors
- servers in businesses and organizations.
- power: 2,500 – 250,000 MFLOPS
- IBM, HP, Dell, Fujitsu, and Sun Microsystems.

(COST: $4,000 – $1,000,000)

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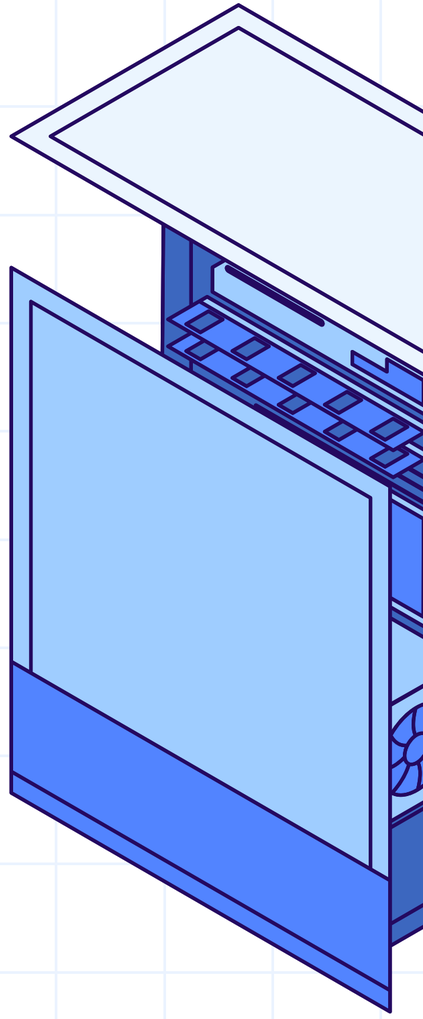
MAINFRAME
Computers



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Mainframes

- are the **heart** of the **computing systems.**
- are a type of computer designed for high throughput, which means **processing data** as **fast as possible**

Key Strengths:

- Processing power ranges from **2,500 to 1,000,000 MFLOPS** (million floating-point operations per second)
- Mainframes are **versatile**. They can manage:
  - Batch and Online Processing
  - Corporate Data Warehouses
  - Web Serving & Network Control

- Cost ranges between **$500,000** to **$20,000,000**.

FIGURE 2.7 IBM System z10 (Courtesy of IBM Corporation)

IBM

- dominant vendor of mainframe computers.
- creator of **IBM System z10.**

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IBM System z10

Symmetric Multiprocessing (SMP):
SMP is used on quad-core processor chips to handle multiple different tasks simultaneously

Massive Capacity:
Up to 64 quad-core processors & 1,520 GB RAM

High Efficiency:
1 z10 can replace 1,500 standard x86 servers

Parallel Sysplex:
Links up to 32 mainframes into 1 unified system (2,048 processors max)

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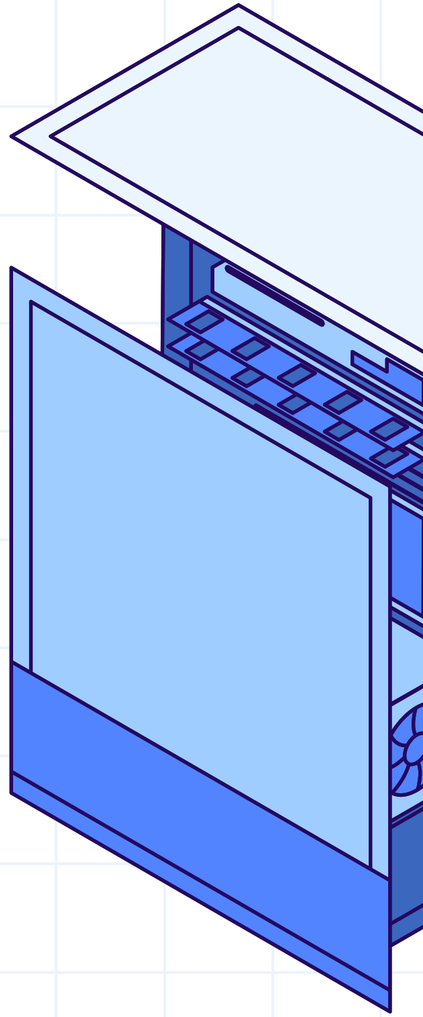
Supercomputers



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Supercomputers

- The Ultimate "**Number-Crunchers**".
- Designed for processing **massive mathematical calculations** at **extreme speeds**.

Key Strengths:

- Performance was rated historically in **MFLOPS** (million floating-point operations per second)
- Today, it is reaching speeds up to **1 Petaflop** (1 quadrillion calculations per second) or higher.

- Cost ranges between **$1 Million** to **$100 Million.**

Cray Inc.

- the acknowledged leader in the high-end supercomputer arena.
- creator of "**Jaguar**", the World's Fastest Supercomputer.

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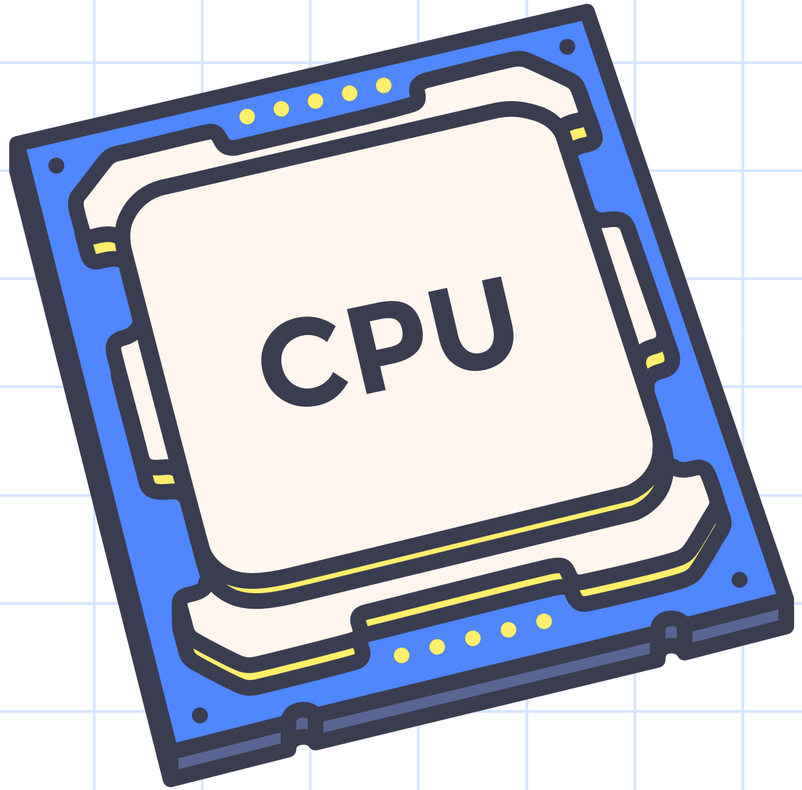
Types of Supercomputers

Type	Main Users	Primary Applications
High-End Supercomputers	Research scientists (Chemists, Physicists, Meteorologists)	Scientific simulations, weather forecasting, advanced research
Midrange Supercomputers	Large business corporations	Research & Development (R&D), massive Web serving, data mining, server consolidation

Supercomputers: How They Work

Parallel Processing

is a multiprocessor configuration (multiple processors installed as part of the same computer system) designed to give a separate piece of the same program to each of the processors so that work can proceed in parallel on the separate pieces.



Three Main Design Structures

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1. Parallel Vector Processing

- Specialized processors designed to run calculations on entire arrays of data simultaneously.

2. Massively Parallel Processing (MPP)

- Thousands of independent processors working together on a single task.

3. Symmetric Multiprocessing (SMP)

- Multiple processors sharing a single main memory pool.

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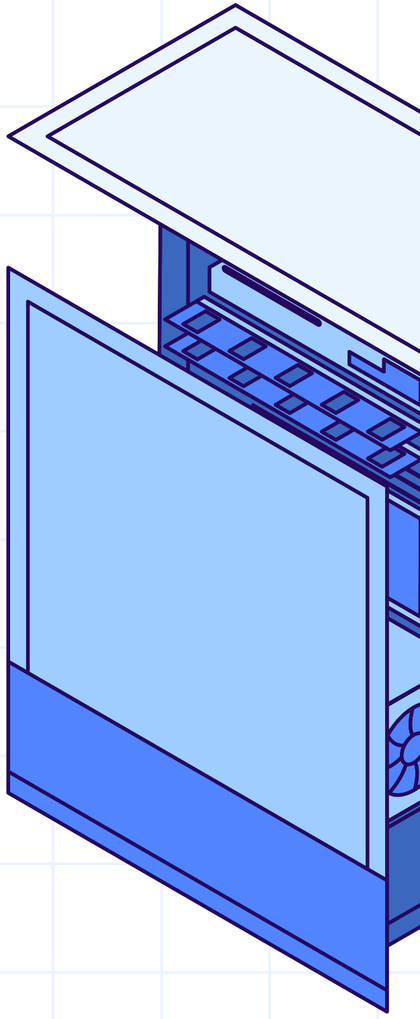
Key Types of SOFTWARE



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Two Major Categories of Software

Applications Software

Support Software

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Software Iceberg

Visible to users

Invisible to users

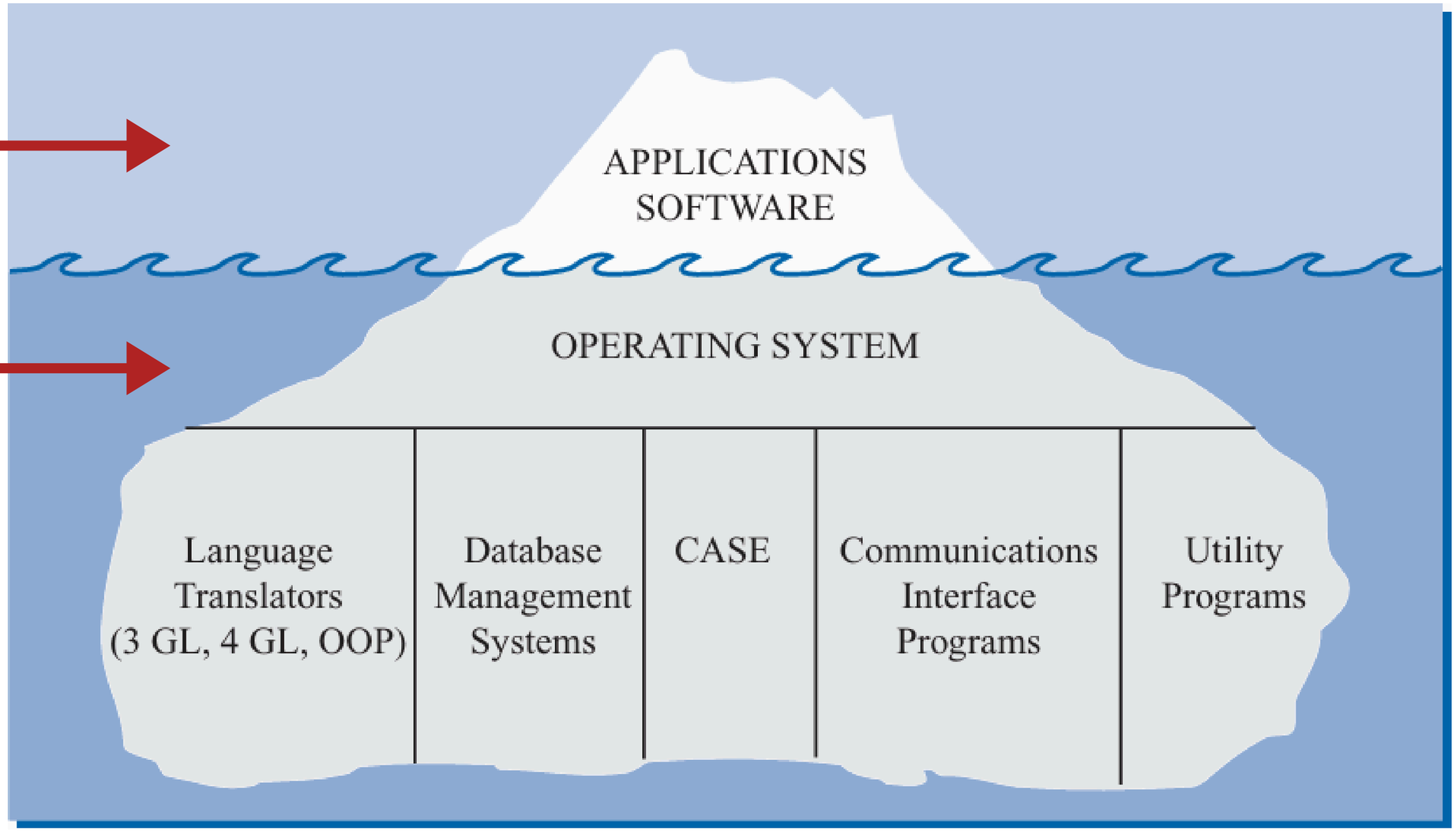

FIGURE 2.8 The Software Iceberg

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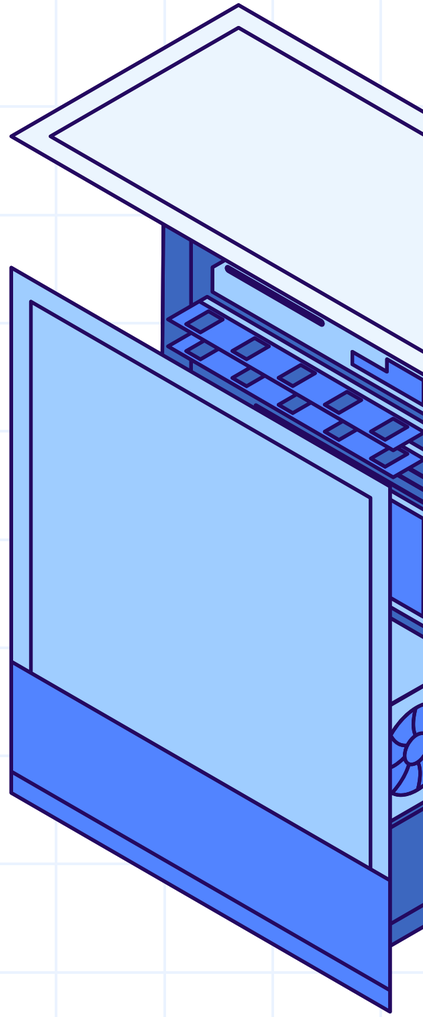
Application SOFTWARE



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Applications Software

- Includes all programs written to accomplish particular tasks for computer users.

Function:

- Produces output that users need to do their jobs.
- Users are often involved in developing or obtaining these programs to meet organizational needs

Examples:

- Portfolio management
- General ledger accounting
- Sales forecasting
- Desktop publishing
- Word processing, spreadsheets

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Sources of Application Softwares

Standard Applications

- **Always purchased**
- Standard products common to all organizations

Examples:

Word, Excel, Email

Unique Applications

- **Always build-in**
- One-of-a-kind applications specific to an organization

Examples:

a unique production control system

Middle Ground Applications

- **Purchased/build in**
- Applications similar across organizations but with specific individual features

Examples:

accounts payable or receivable, general ledger, inventory and etc.

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User Application Development:

- Built by **computer-oriented employees** taking advantage of desktop computers and easy-to-use tools.
- Primarily used for smaller, personal, or departmental systems.

IS Organization Development:

- Built by **Official IS departments**, outside consulting firms, or specialized software vendors.
- IS teams use both end-user tools and advanced development tools to create company-wide software systems that the whole organization uses.

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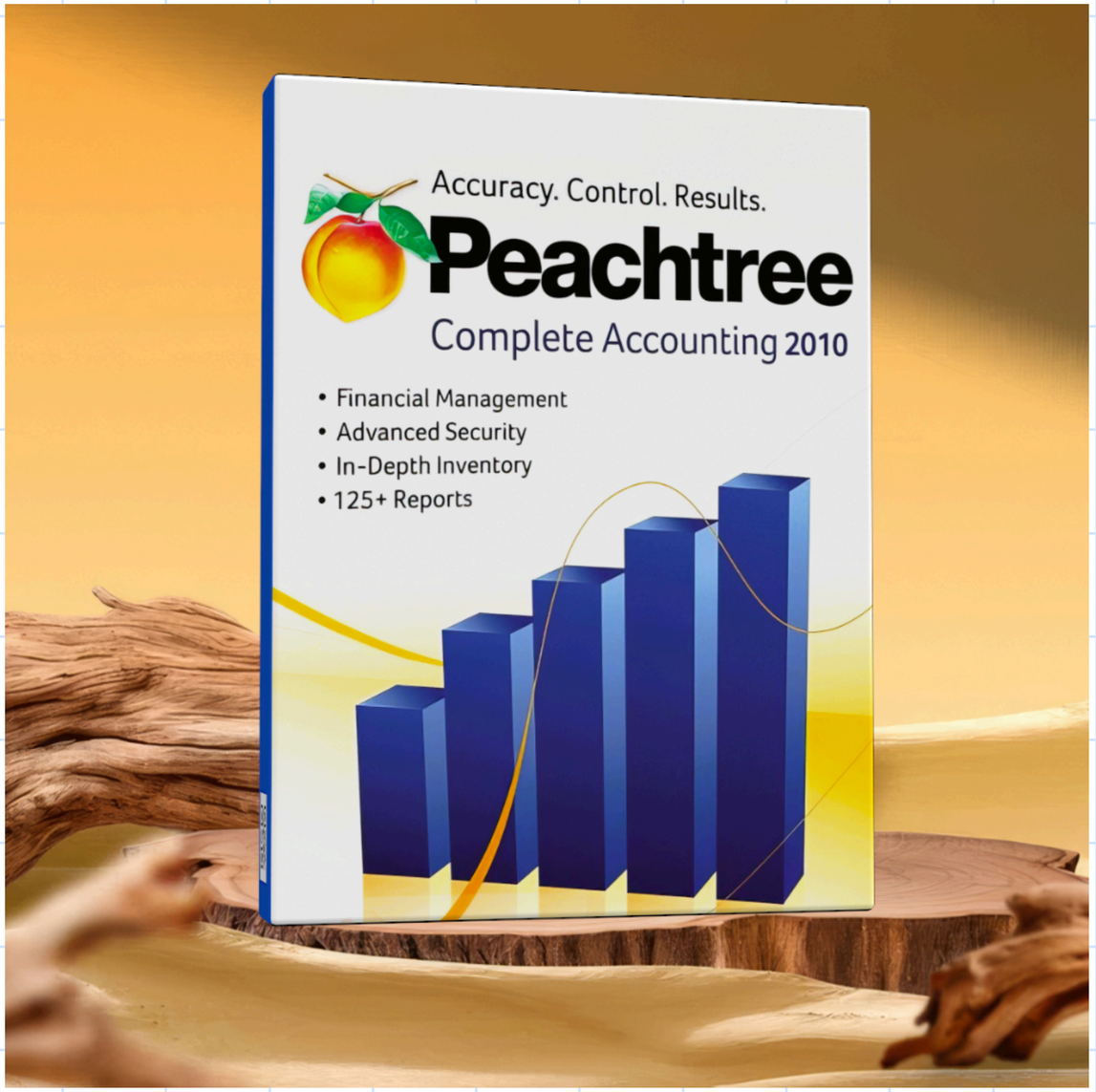
Peachtree by Sage Premium Accounting 2010

Its a commercial **software package** for small to midsized businesses with a retail price of $500 for a single-user version.



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Personal Productivity Software

- It is **microcomputer-based** application software developed with a friendly, comfortable **Graphical User Interface (GUI).**

WHAT DOES IT DO?

refers to application programs designed to allow individuals to create, analyze, communicate, and manage information efficiently on personal devices.

Why is this market so successful for businesses today?

- Affordable Pricing
- Product Reviews
- Continuous Innovation

VisiCalc (1979)

- the very first electronic spreadsheet

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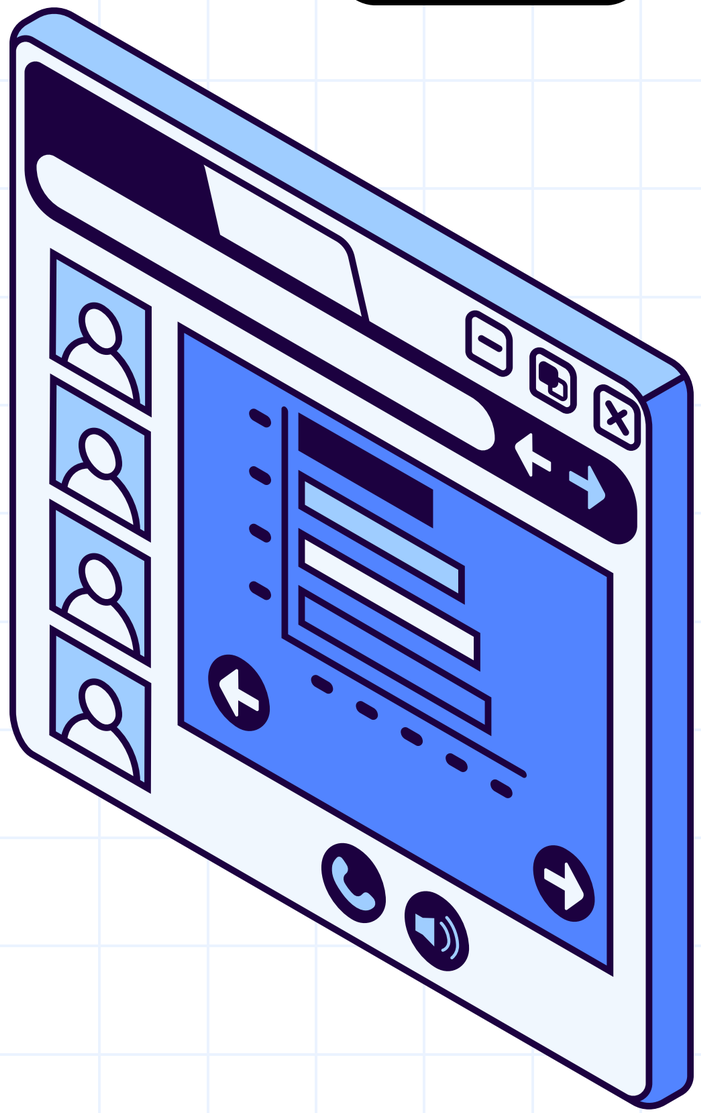
The Core Categories of PERSONAL PRODUCTIVITY SOFTWARE



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1 Word Processing

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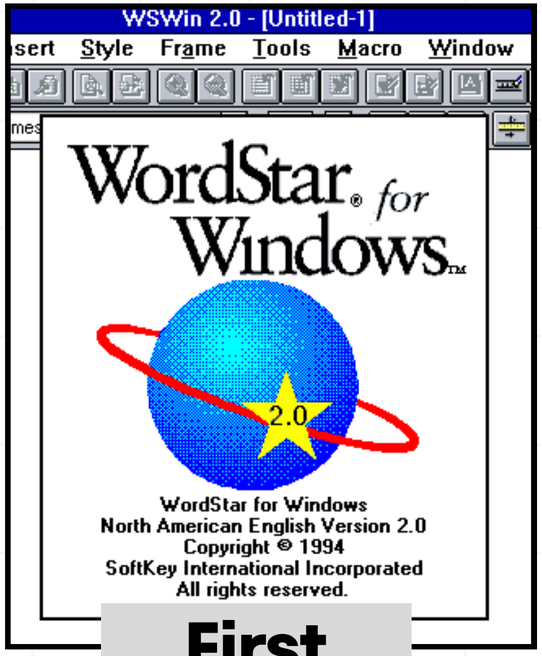
Definition: is perhaps the **most widespread/ubiquitous type** of personal productivity software.

WordStar

WordPerfect

MS Word



First

1980s

1990s

Key Features:

- Underlines potential spelling errors for real-time correction.
- Allows easy adjustments to fonts, margins, and columns.
- Rewrites sentences for grammatical correctness with a single mouse click.
- Converts Web files directly into usable Word format.

1 Word Processing

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StarOffice Writer

OpenOffice Writer

Google Docs

2 Spreadsheets

NEXT PAGE

Definition: Electronic spreadsheets are **second only** to word processing in popularity among personal productivity software.

VisiCalc

Lotus 1-2-3

MS Excel

First

1980s

Current

Accountant's Spreadsheet

large piece of paper divided into rows and columns to organize financial data.

Applications:

- Sales forecasts for the next 12 months
- An instructor's gradebook
- Budget summaries across multiple time periods

2 Spreadsheets

NEXT PAGE

Corel Quattro Pro

Sun’s StarOffice Calc

IBM Lotus Symphony Spreadsheets

3 Database Management Systems (DBMSs)

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Definition: microcomputer-based database management systems (DBMSs) are the **third most popular category** of personal productivity software.

dBase

MS Access

1980s

Current

Useable Tools:

- **Macros**
- **Programming Tools**

Core Model:

- Relational Data Model
- Desktop DBMSs are more simple

4 Presentation Graphics

NEXT PAGE

Definition: Used for creating **largely textual presentations**, but with embedded clip art, photographs, graphs, and other media.

MS Powerpoint

- is the leader in this field.

Spreadsheets vs. Specialized Tools

Specialized presentation graphics products offer **even greater** capabilities.

Specialized Tools for Complex Graphics

Visio

Adobe Illustrator

CorelDRAW GRAPHICS SUITE

5 Electronic Mail and Groupware

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Electronic Mail

Definition:
Has become the **preferred way of communicating** for managers in most businesses today.

Key Characteristics:

- Asynchronous
- Unobtrusive
- Easy to use
- Precise

Groupware

Definition:
Incorporates electronic mail, but includes **much more.**

Primary Goal:

- To help a group of people become more productive.

Key Features:

- Provides innovative ways of data sharing.

6 Office Suites

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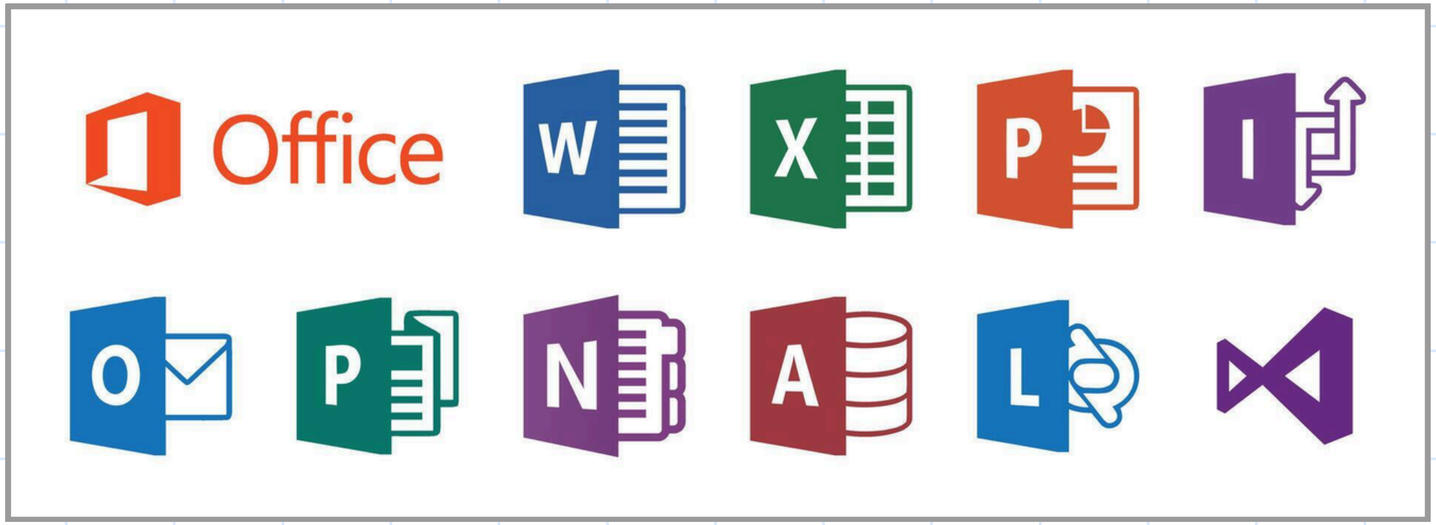
Definition: Packages that **combine** certain personal productivity software applications **into integrated suites for office use.**



Microsoft Office

- the market leader which was introduced in 1990.

Reasons for its dominance:

- First-mover advantage
- Control of the OS
- Good individual products
- Better integration of individual products compared to competitors

Alternatives:

7 World Wide Web Browsers

NEXT PAGE

Definition: is the software that runs on the user's microcomputer, enabling the user to look around, or "**browse**," **the Internet.**

- Connected to the Internet
- Browsers use **Hypertext** to navigate.

ISP (Internet Service Provider)

LAN (Local Area Network)

Hypertext

- this is the **linking of objects**, such as text, pictures, sound clips, and video clips, to each other so that by **clicking on highlighted text** or a small icon, the user is **taken** to the related object.

- http://www.whoi.edu/science/B/people/kamaral/hypertext.html

- http://www.cyberartsweb.org/cpace/cpace/ht/systems.html

7 World Wide Web Browsers

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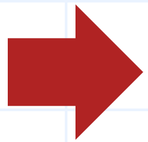
Netscape
(Mid 1990s)



Internet Explorer
(Late 1990s & 2000s)

Modern Competitors

mozilla Firefox®

Google Chrome

Opera

Safari

Web Delivery Models

Technology	Who Starts It?	Best Example
Pull Technology	You request it	Browsing a website by clicking a link, **RSS** (Really Simple Syndication) **Reader**
Push Technology	The Server sends it	Getting an email or an automated system update

7 Other Personal Productivity Products

NEXT PAGE

Software Category	Core Purpose / Function	Product Examples
Desktop Publishing	Design and print professional in-house publications (brochures, magazines, reports).	Adobe InDesign, MS Office Publisher, QuarkXPress,
Computer Security	Protect systems and data from cyber threats	Norton Internet Security, ZoneAlarm Extreme Security
Contact Management	Track customer data and interaction history	ACT!, GoldMine, Up-to-Date & Contact

7 Other Personal Productivity Products

NEXT PAGE

Software Category	Core Purpose / Function	Product Examples
Project Scheduling	Plan, track, and manage project timelines and tasks	Microsoft Office Project, Project KickStart Pro
Image Editing	Enhance, edit, and manipulate digital photos	Adobe Photoshop Elements, Corel Paint Shop Pro Photo
Video Editing	Cut, edit, and produce digital video projects	Adobe Premiere Elements, Corel VideoStudio
System Utilities	Perform essential maintenance tasks (disc burning, data backup)	Roxio Creator, Roxio BackOnTrack Suite

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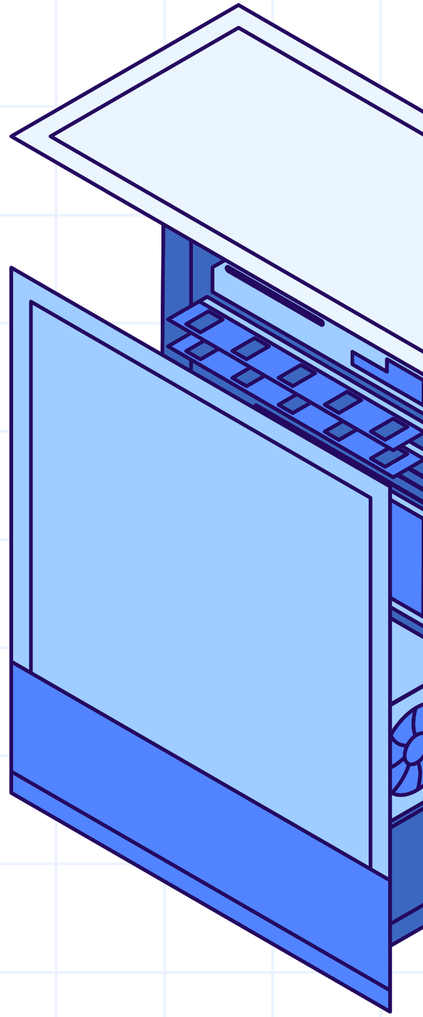
Support
SOFTWARE



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Support Software

- has been designed **to work in the background** to ensure application programs (like Word, Chrome, or Excel) **run smoothly**.

Function:

- Does not directly produce user-facing output.
- Instead, it creates the environment necessary for application software to function.

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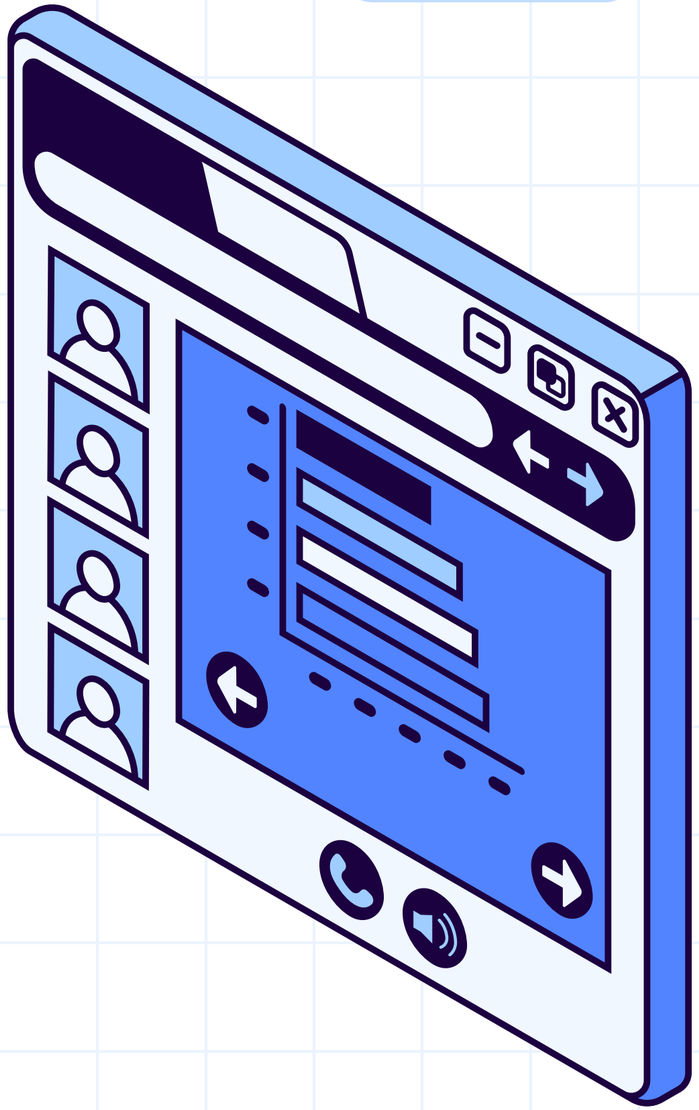
The Various Types of SUPPORT SOFTWARES



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Operating System

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A complex support program that **controls** computer hardware and **coordinates** all other software.

- Acts as an intermediary
- Originated in the mid-1960s

Key Functions of an OS:

- Job & Queue Management
- Network Communications
- File & Library Management

Main Purpose:

- **Maximize Throughput:** Keeps CPU running at speed to get more work done with the available resources.
- **Ease User Workload:** Creates a user-friendly environment that makes it easy to run programs and manage files.

Operating System

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How Users Interact with the OS

System Type	Method	Interaction
PC or Mac	Graphical Interface	Users click icons or labels to run programs or retrieve files.
Large Machines (Mainframes)	Job Control Language (JCL)	Users enter specific text instructions defining program names, required data files, output rules, and account numbers.

Advanced Operating Systems Concepts

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Multiprogramming

- Used on large computers.
- **Switches** between programs in memory to overlap input and output with processing time.

Multitasking

- Used on micro-computers.
- Same function as multiprogramming, applied to **smaller machines**.

Concepts to Increase CPU Efficiency

Multithreading

- Thread is short for **"thread of execution."**
- Switches between different parts (threads of execution) of the same program near simultaneously.

Advanced Operating Systems Concepts

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Virtual Memory

- **Manages main memory**, making memory seem unlimited so programs can exceed physical cell limits.

- Only active program segments execute; inactive parts stay on DASDs (storage devices).
- Programs are split into pages.
- OS switches pages back and forth between main memory and DASDs.

Multiprocessing

- Occurs when **two or more CPUs** are installed in one computer system.

- Each CPU runs its own jobs while one OS controls and tracks all CPUs.

Sources of Operating Systems

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Proprietary Systems

Tied directly to specific hardware or vendor ecosystems.

Open Systems

Systems that are **not tied** to a specific computer system or hardware manufacturer.

Open Systems

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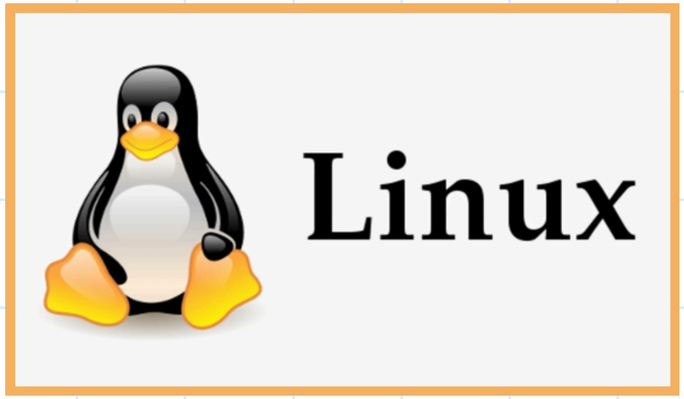
UNIX

- Created in the 1960s at Bell Labs
- Usually commercial / expensive
- Large corporate mainframes & high-end servers
- Modifying it requires corporate licensing



LINUX

- Created in 1991 by Linus Torvalds
- Free to download, use and modify.
- Millions of servers, supercomputers, PCs, and Android phones
- If you make improvement, you must share them back.

Sources of Operating Systems

NEXT PAGE

Server Operating Systems

- Also called **Network Operating Systems (NOS)**
- Software installed on a server that manages network resources
- Allows connected computers to share resources.

IT Platform

- The set of hardware, software, communications, and standards an organization uses to build and operate its information systems

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Language TRANSLATORS

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Language Translators

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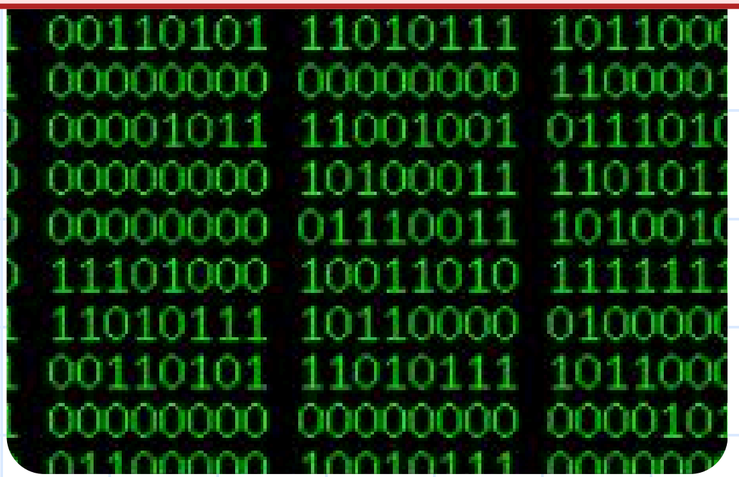
Machine Language (1GL)

- which is specific to a particular computer model.
- do not have to be translated
- may be run directly on the computer model for which they were written.

- very tedious and demanding for humans to write



created human-like languages
along with
language translators

Generations of Computer Languages

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Machine Language - First Generation Language (1GL)

- The native language of a specific computer model.
- Does not require translation to run.

Assembly Language - Second Generation Language (2GL)

- Created to automate tedious tasks
- Uses **mnemonic operation codes**
- Uses **symbolic addresses**
- Uses an **Assembler**

- translates the Assembly Language program into Machine Language.

Later Generations

- Third-Generation (3GLs)
- Fourth-Generation (4GLs), and
- Object-Oriented Programming (OOP) languages

Language Translators: The Process

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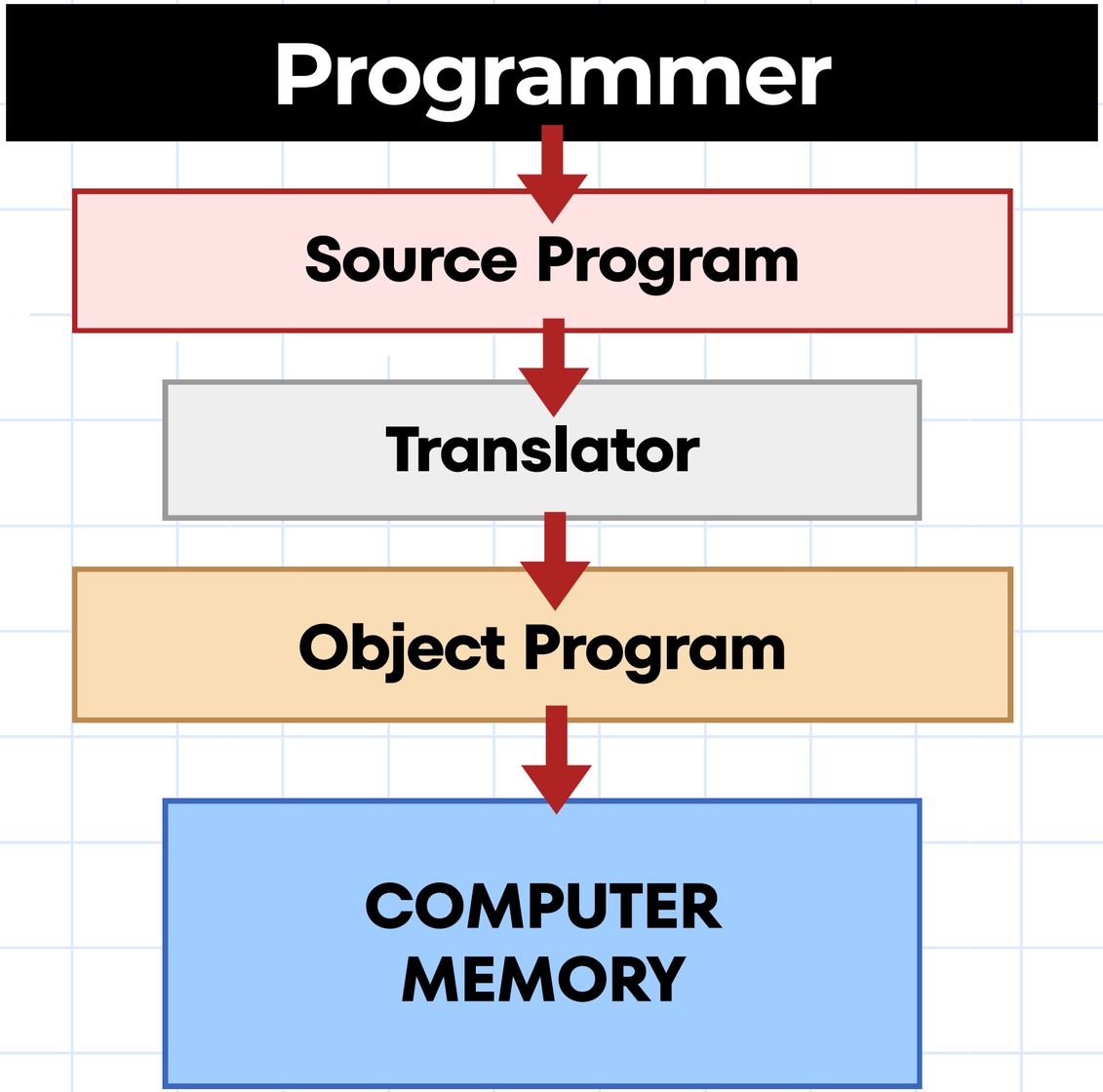

The code before translation

The code after translation (machine language).

Third Generation Languages (3GLs)

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What are they?

- **Human-Designed:** Designed strictly for humans to easily learn and use.
- **Procedural Languages -** Programmers write step-by-step procedures to accomplish a task.
- **One-to-Many Translation.**
- **Machine Independence:** Can run on many different types of computers.

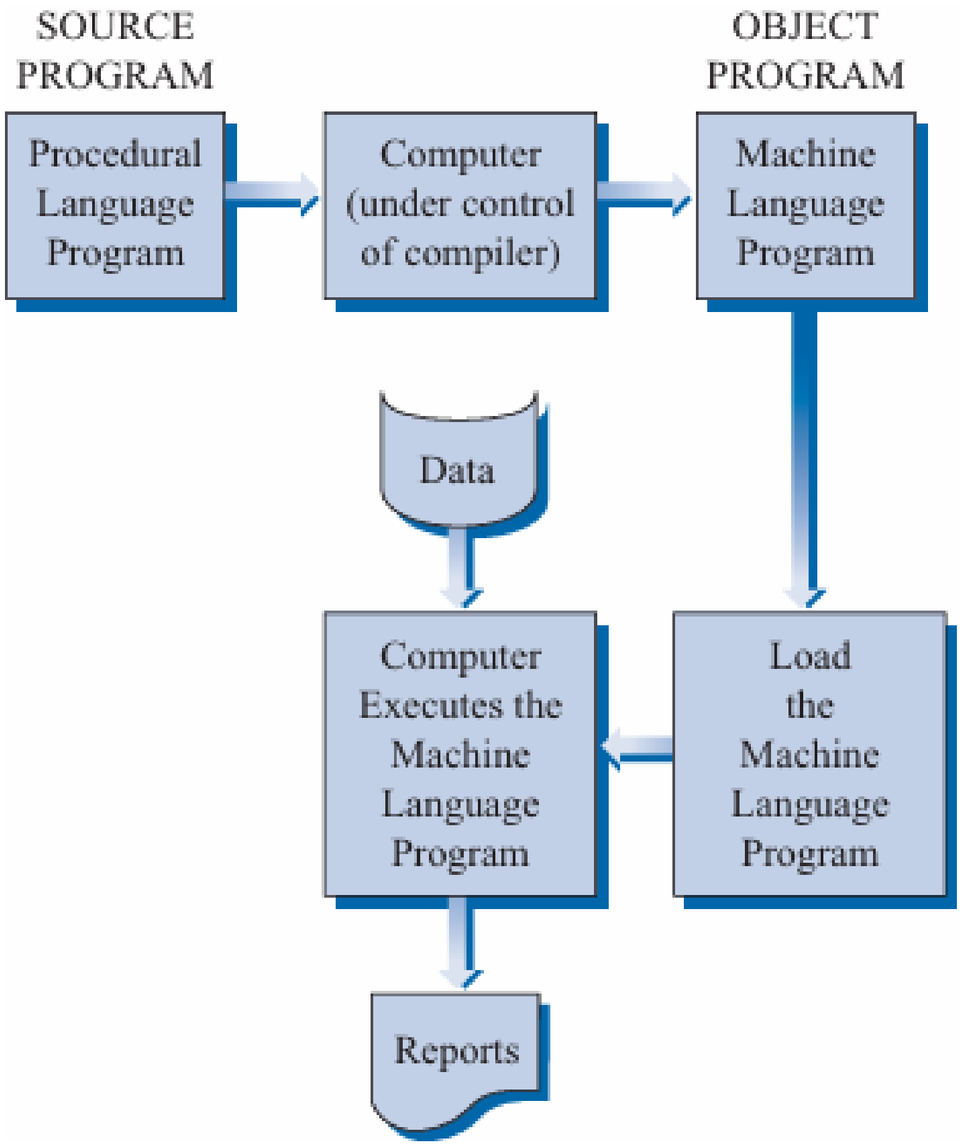
the **earliest** procedural language

Examples:

- **COBOL** - Common Business-Oriented Language
- **BASIC** - Beginner's All-purpose Symbolic Instruction Code
- **C**

Third Generation Languages (3GLs)

NEXT PAGE



How are they translated?

Compiler

- Translates the entire program into machine language all at once before any execution starts.

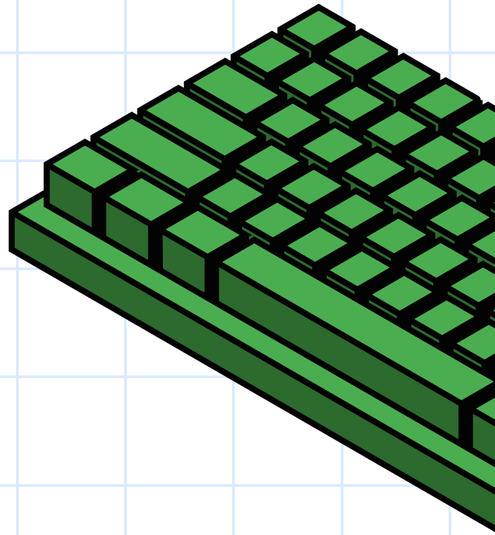
Interpreter

- Translates the program one statement at a time.

COBOL

NEXT PAGE

- COBOL stands for **CO**mmon **B**usiness-**O**riented **L**anguage.
- Introduced in **1960**
- Provides an **industry-wide common language.**
- Developed by the COBOL Committee of CODASYL.
- The **most popular language** for programming mainframe computers for business applications.
- Known as **verbose**.



The Four Divisions of a COBOL Program

- Identification Division
- Environment Division
- Data Division
- Procedure Division

```
000100   IDENTIFICATION DIVISION.
000200   PROGRAM-ID. HELLO1.
000300   ENVIRONMENT DIVISION.
000800   CONFIGURATION SECTION.
000900   SPECIAL-NAMES.   SYSIN IS
001000   INPUT-OUTPUT SECTION.
001100   DATA DIVISION.
001200   FILE SECTION.
001300   WORKING-STORAGE SECTION.
001400   01   WORK-FIELDS.
```

COBOL

NEXT PAGE

```
1      8   12
       IDENTIFICATION DIVISION.
       PROGRAM-ID. COMMISSIONS-COMPUTE.
       ENVIRONMENT DIVISION.
       CONFIGURATION SECTION.
       SOURCE-COMPUTER. IBM-AS400.
       OBJECT-COMPUTER. IBM-AS400.
       INPUT-OUTPUT SECTION.
       FILE-CONTROL.
           SELECT SALES-FILE ASSIGN DISK-FILEIPT.
           SELECT COMMISSIONS-FILE ASSIGN DISK-FILERPT.
       DATA DIVISION.
       FILE SECTION.
       FD SALES-FILE
           LABEL RECORD OMITTED
           RECORD CONTAINS 80 CHARACTERS
           DATA RECORD IS IN-RECORD.
       01  IN-RECORD                PICTURE X(80).
       FD  COMMISSIONS-FILE
           LABEL RECORD OMITTED
           RECORD CONTAINS 132 CHARACTERS
           DATA RECORD IS PRINT-RECORD.
       01  PRINT-RECORD             PICTURE X(132).
       WORKING-STORAGE SECTION.
       01  SALES-RECORD.
           05  NAME                 PICTURE A(30).
           05  FILLER               PICTURE X(10).
           05  SALES                PICTURE 9(8)V99.
           05  FILLER               PICTURE X(30).
       01  COMMISSION-RECORD.
           05  FILLER               PICTURE X(10).
           05  NAME-OUT             PICTURE A(30).
           05  FILLER               PICTURE X(10).
           05  SALES-OUT            PICTURE $$$,$$$,$$$.99.
           05  FILLER               PICTURE X(10).
           05  COMMISSION           PICTURE $$$$,$$$.99.
           05  FILLER               PICTURE X(47).
       77  TEMP-COMMISSION          PICTURE 9(6)V99.
       77  TOTAL-COMMISSIONS        PICTURE 9(10)V99  VALUE 0.
       77  TOTAL-COMM-EDITED        PICTURE $$,$$$,$$$,$$$.99.
       01  MORE-DATA                PICTURE X         VALUE 'Y'.
           88   THERE-IS-MORE-DATA                    VALUE 'Y'.
           88   THERE-IS-NO-MORE-DATA                 VALUE 'N'.
```

```
1      8   12
       PROCEDURE DIVISION.
       MAIN-CONTROL.
           PERFORM INITIALIZATION.
           PERFORM READ-PROCESS-PRINT UNTIL THERE-IS-NO-MORE-DATA.
           PERFORM COMPLETE.
           STOP RUN.
       INITIALIZATION.
           OPEN INPUT SALES-FILE, OUTPUT COMMISSIONS-FILE.
           MOVE SPACES TO COMMISSION-RECORD.
       READ-PROCESS-PRINT.
           READ SALES-FILE INTO SALES-RECORD
               AT END MOVE 'N' TO MORE-DATA.
           IF THERE-IS-MORE-DATA
               MOVE NAME TO NAME-OUT
               MOVE SALES TO SALES-OUT
               IF SALES GREATER 50000
                   COMPUTE TEMP-COMMISSION = .01*50000+.02* (SALES-50000)
               ELSE
                   COMPUTE TEMP-COMMISSION = .01*SALES
               MOVE TEMP-COMMISSION TO COMMISSION
               WRITE PRINT-RECORD FROM COMMISSION-RECORD
                   AFTER ADVANCING 1 LINES
               ADD TEMP-COMMISSION TO TOTAL-COMMISSIONS.

       COMPLETE.
           MOVE TOTAL-COMMISSIONS TO TOTAL-COMM-EDITED.
           DISPLAY 'TOTAL-COMMISSIONS ARE' TOTAL-COMM-EDITED.
           CLOSE SALES-FILE, COMMISSIONS-FILE.
```

Other Procedural Languages

NEXT PAGE

FORTRAN

Mid 1950s

- Developed for scientific software

- "Grand-daddy" of procedural languages.
- Built for scientific calculation

BASIC

Beginner's All-purpose Symbolic Instruction Code

Early 1960s

- John Kemeny & Thomas Kurtz (Dartmouth College)

- Simplest procedural language.
- Easy-to-learn and interactive.
- Designed for students to focus on logic.

C

1970s

- Dennis Ritchie & Brian Kernighan

- Highly powerful, but harder to use.
- Closer to assembly language and less English-like.
- Originally used to write the UNIX operating system.

Fourth Generation Languages

- Also known as **Productivity Languages** or **Nonprocedural Languages**
- Easier to use than Third Generation Languages (3GLs)
- Users tell the computer what they want to do instead of how to do it
- Uses **English-like syntax**, making programming simpler

Castañeda, Chayris Angel M.

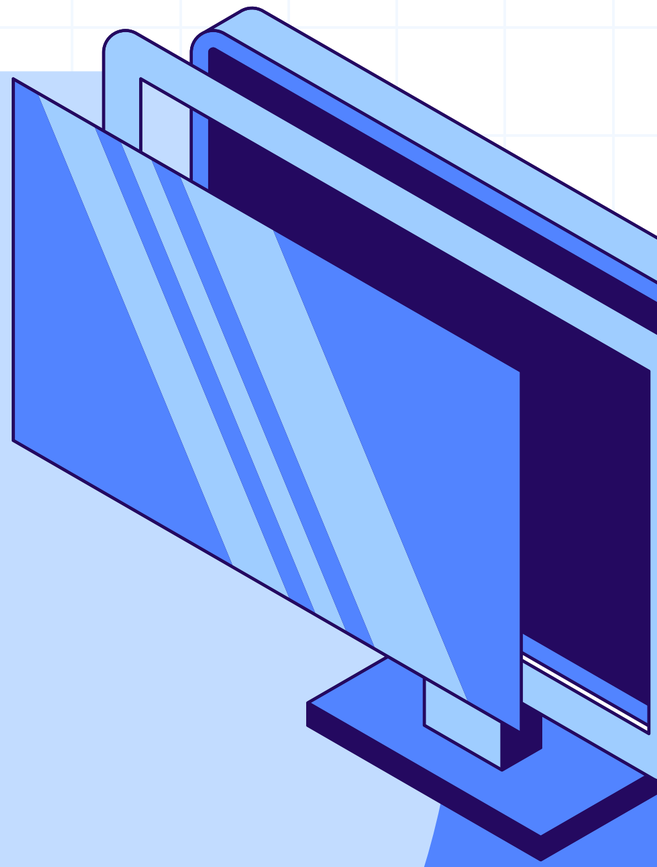
Advantages and Disadvantages of 4GLs

Advantages

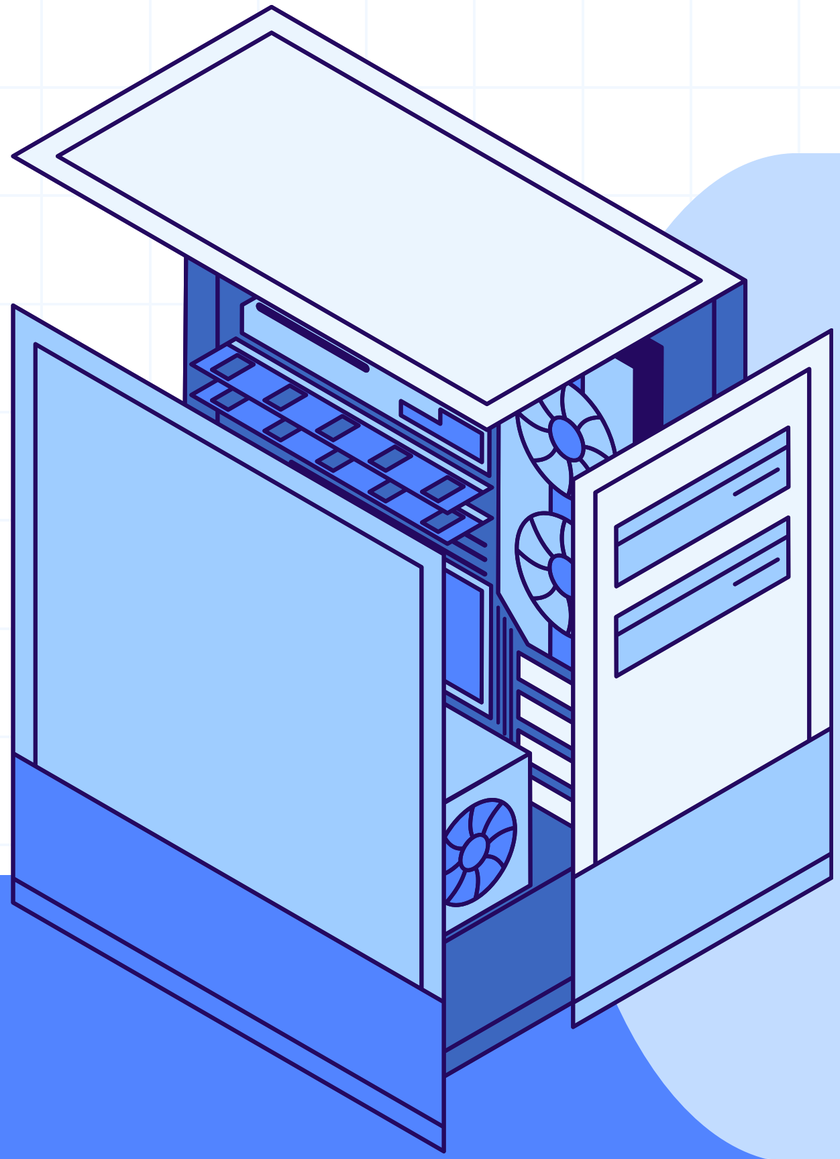
- Requires fewer lines of code
- Easier to write and understand
- Easier to modify and maintain
- Less prone to programming errors
- No need to manage memory manually

Disadvantages

- Not suitable for all types of applications
- Usually slower than 3GL programs
- Uses more computer resources
- Less efficient for programs that run frequently



Application of 4GLs



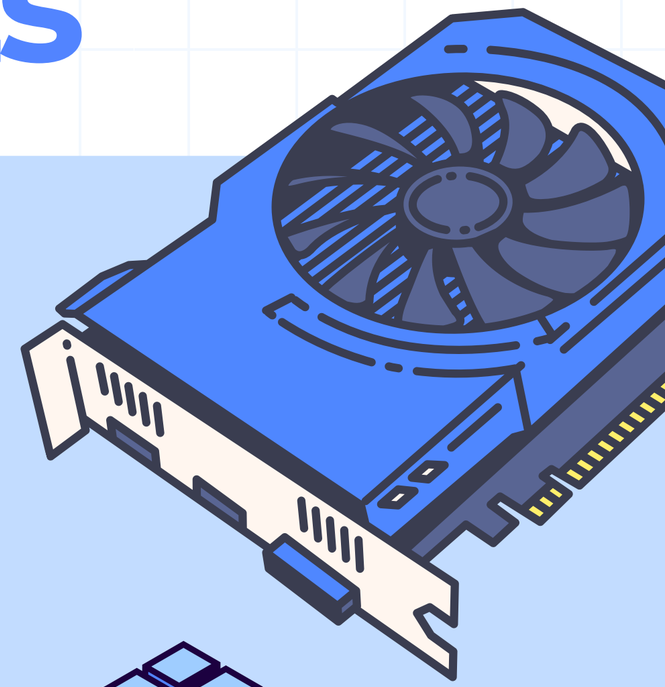
Common Uses

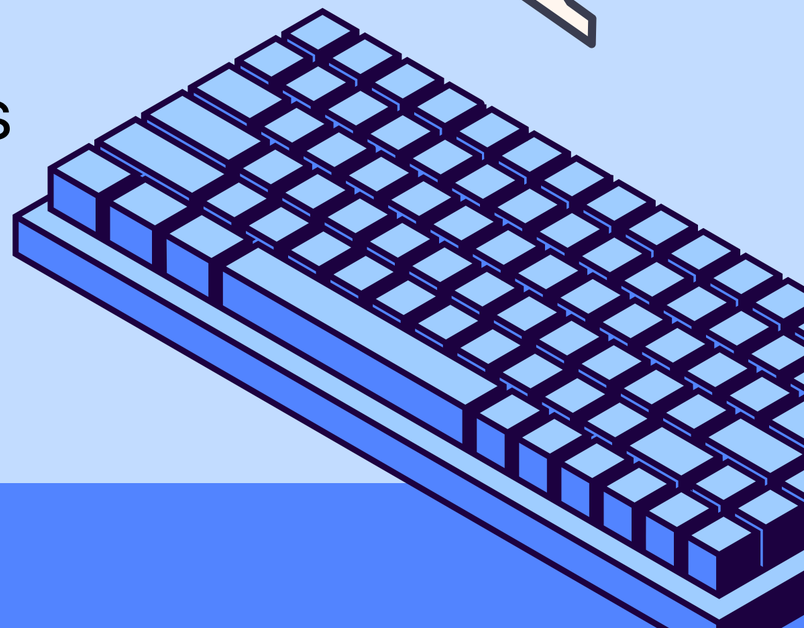
- Decision Support Systems (DSS)
- Business reports
- Financial modeling
- Statistical analysis
- Data management

Evolution of 4GLs

Evolution

- Introduced in 1967 with RAMIS
- FOCUS became another popular 4GL
- Later evolved into Business Intelligence (BI) tools
- Focus shifted toward analyzing business data



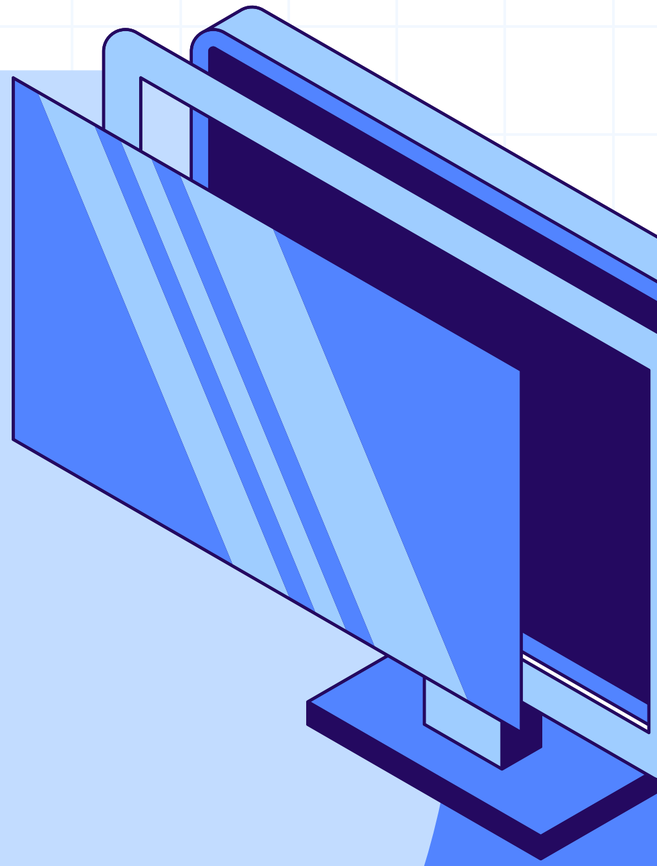


Examples of Modern BI Tools

Examples

- SAS Business Intelligence
- WebFOCUS
- IBM Cognos
- SAP BusinessObjects
- Oracle Business Intelligence
- Microsoft SQL Server Analysis & Reporting Services

Future of 4GLs



Future Trends

- Continued growth of Business Intelligence
- Better graphical reports and dashboards
- More user-friendly tools
- Development of **Natural Language Programming**

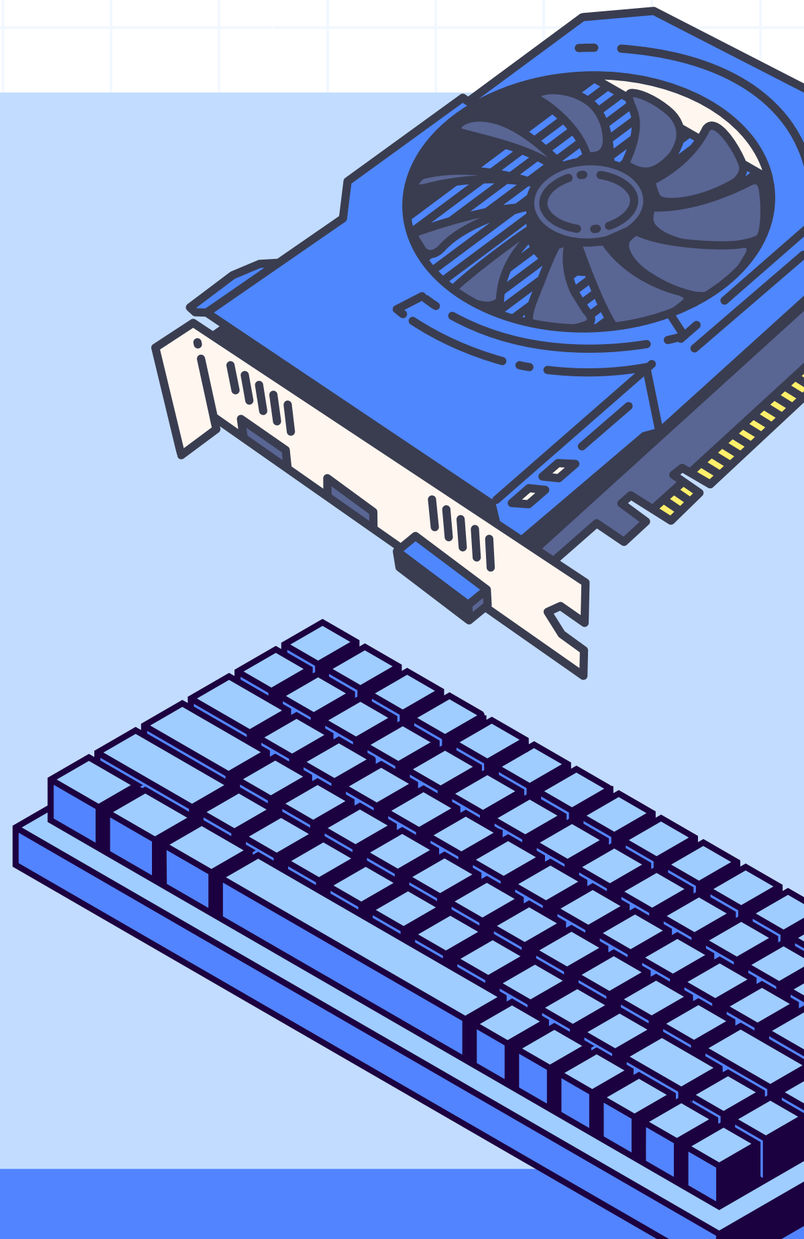
Markup Languages

- Different from **3GLs, 4GLs, and Object-Oriented Programming (OOP) languages**
- Used to **describe or format data**, not to create programs
- Controls how information is displayed or organized
- Commonly used in web development

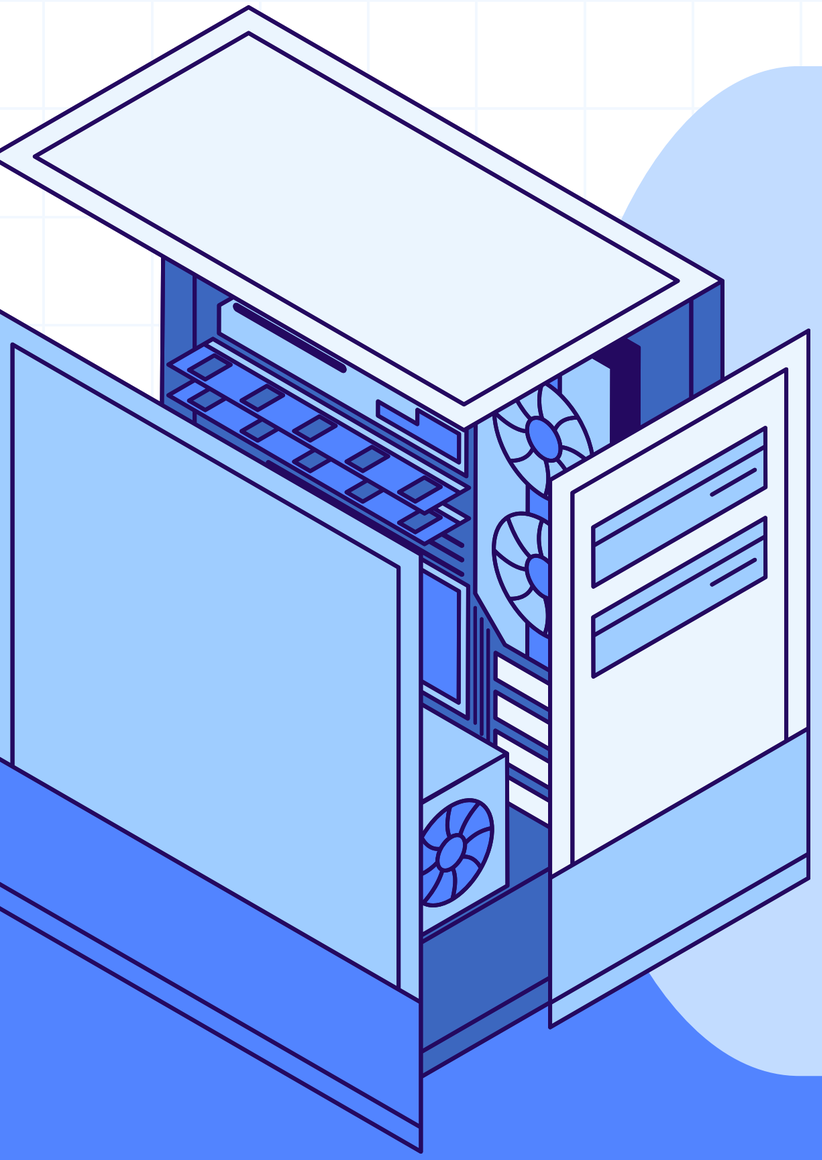
HTML (HyperText Markup Language)

What is HTML?

- Standard language for creating web pages
- Uses **tags** to format content
- Can display:
  - Headings
  - Bold and italic text
  - Images
  - Hyperlinks
  - Videos (HTML5)



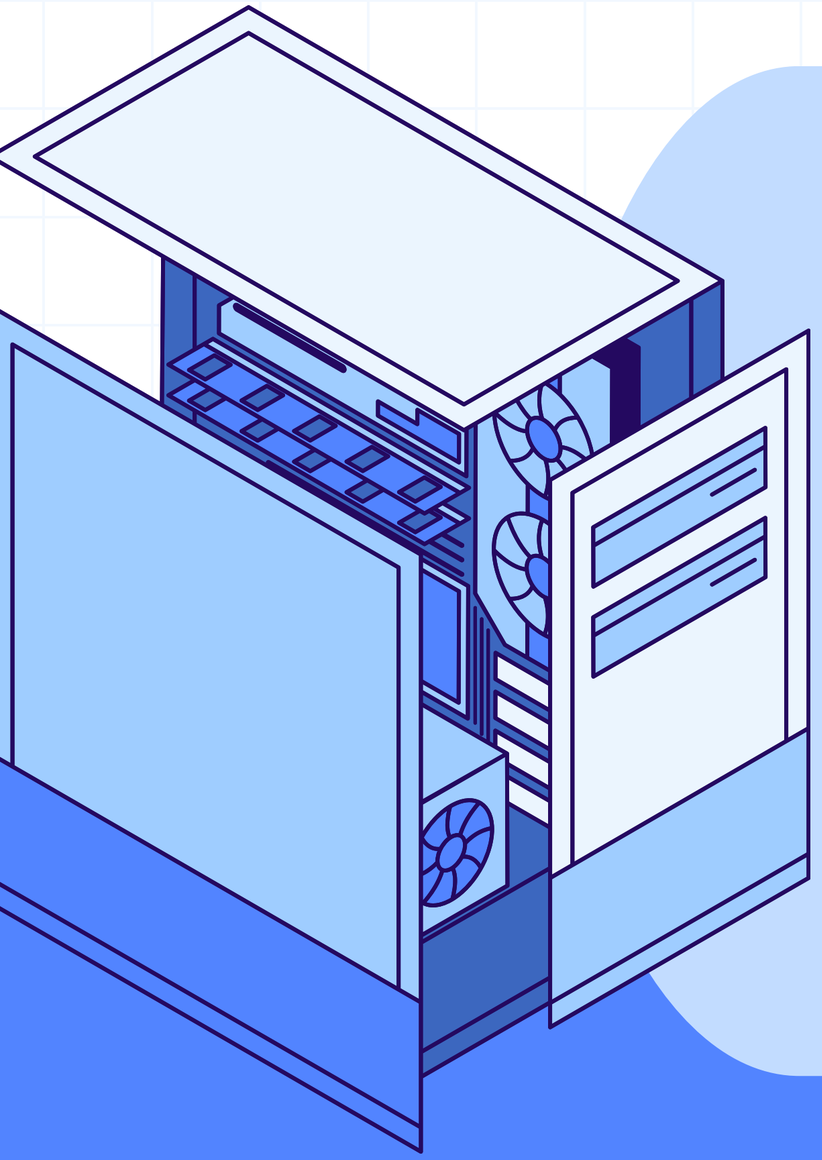
VRML (Virtual Reality Modeling Language)



What is VRML?

- Stands for Virtual Reality Modeling Language
- Used to display 3D objects on the web
- Considered the 3D version of HTML

XML (eXtensible Markup Language)



What is XML?

- Designed to store and exchange data
- Identifies the meaning of data
- Developed by the World Wide Web Consortium (W3C)
- Uses customizable tags

Features of XML

- Extensible (new tags can be created)
- Supports nested tags
- Separates data from presentation
- Enables easy data sharing
- Improves data integration across organizations

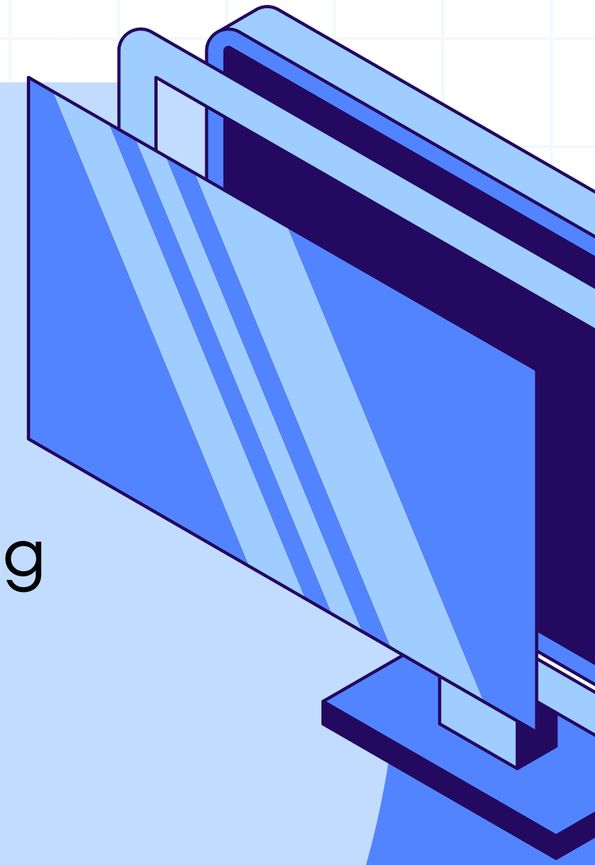
HTML vs XML

HTML

- Displays information
- Focuses on presentation
- Uses predefined tags
- Creates web pages

XML

- Describes information
- Focuses on data meaning
- Uses customizable tags
- Shares data between systems



XSL (eXtensible Style Language)

Purpose of XSL

- Controls how XML data is displayed
- Converts XML into formats like HTML
- Separates data from presentation

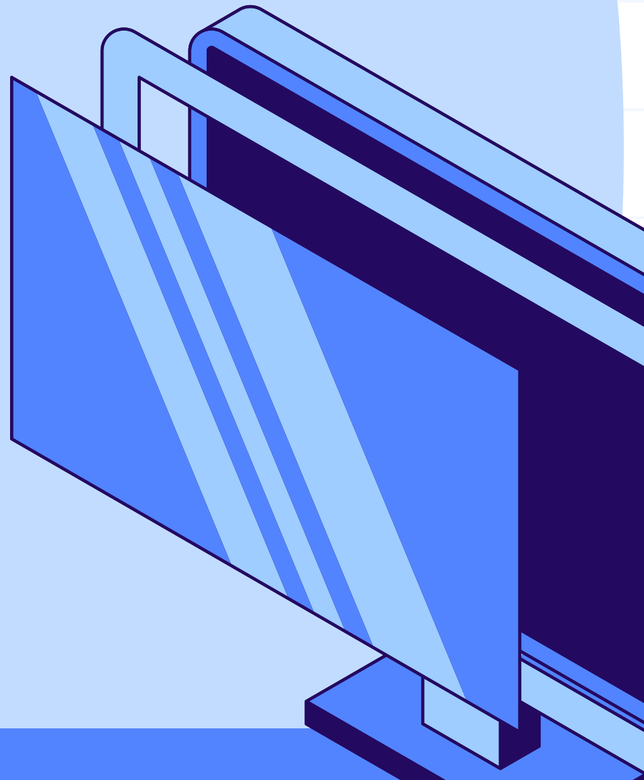
Web Services

What are Web Services?

- Allow different applications to communicate
- Enable software reuse
- Work across different hardware and programming languages
- XML is the key technology that makes this possible

Importance of Web Services

- Faster software development
- Lower development costs
- Easier integration of applications
- Reusable software components
- Better communication between systems



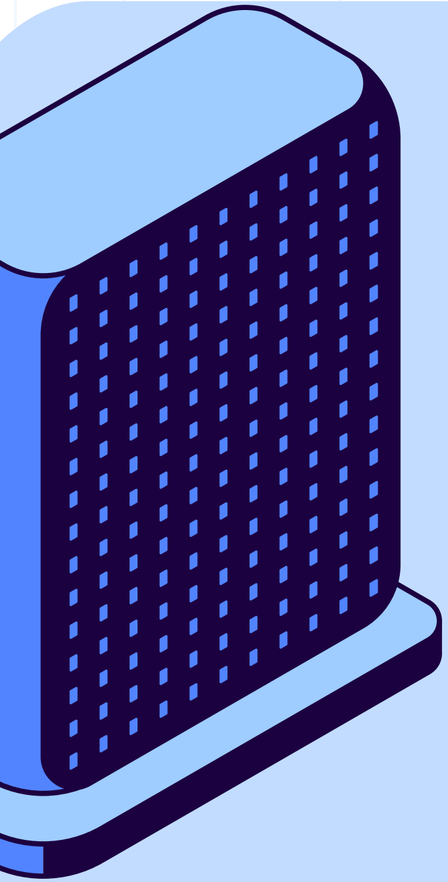
Object-Oriented Programming (OOP)

- A programming paradigm introduced in the 1970s
- Different from both 3GLs and 4GLs
- Focuses on creating and reusing objects
- Makes programs easier to develop and maintain

Key Concepts

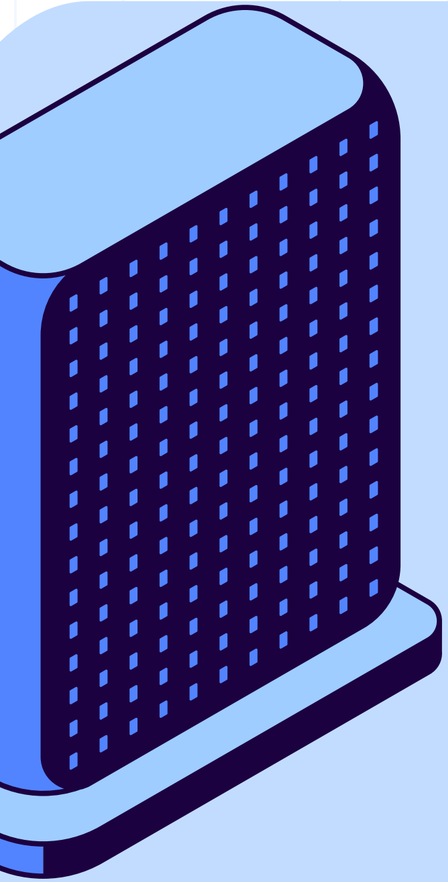
- Create an object once
- Store it for reuse
- Objects can represent:
- User interface components (buttons, text boxes)
- Real-world entities (Employee, Factory, Customer)

Common OOP Languages



- **C++** – Object-oriented version of C
- **Java** – Platform-independent language
- **Visual Basic .NET (VB.NET)** – Microsoft language
- **C# (C Sharp)** – Microsoft language for .NET applications

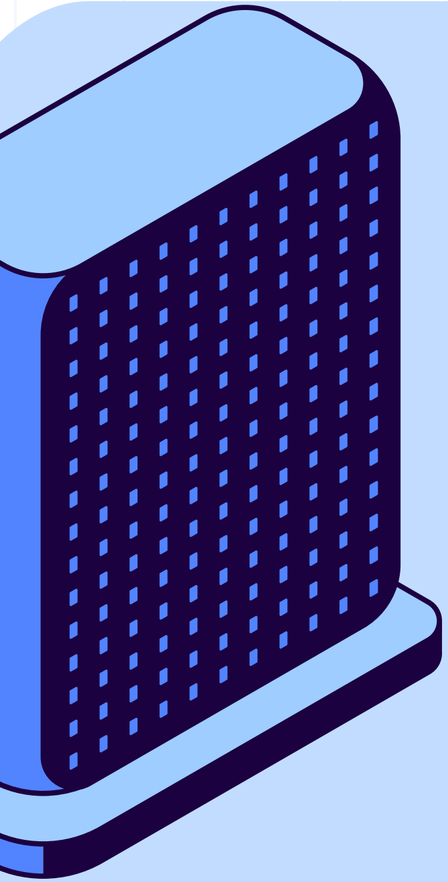
JAVA APPLICATIONS



Types of Java Programs

- **Standalone Applications** – Run on a computer
- **Applets** – Run inside a web browser
- **Servlets** – Run on a web server

OOP Components



- **Class –** Blueprint for creating objects
- **Object** – An instance of a class
- **Properties** – Characteristics or attributes
- **Methods** – Actions an object can perform
- **Events** – Actions that trigger object responses

```
public class Dog{
	double height;
	double weight;
	String color;

		public Dog (double someheight, double someweight, String somecolor)
	{
		height = sometype;
		weight = someweight;
		color = somecolor;
	}

		//methods
		public void sleep() {
		//code to make a dog sleep will go here
		}
		public void run() {
		//code to make a dog run will go here
		}

		public Object fetch() {
		//code to make a dog fetch will go here
		//this method will return the item fetched
		}
	public void display()
	{
	System.out.println("The Height of Animal is: " + height);
	System.out.println("The Weight of Animal is: " + weight);
	System.out.println("The Color of the Animal is: " + color);
	}
}
```

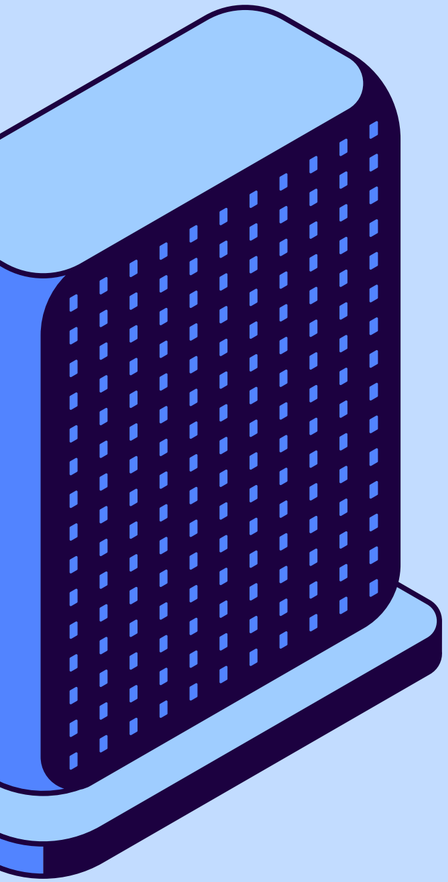
FIGURE 2.12 Java Class Called Dog

```
public class AnimalTest
{
	public static void main(String args[])
	{
		Animal myanimal;

		myanimal = new Dog("10.5", 30,"Black");
		myanimal.display();
	}
}
```

FIGURE 2.13 Java Instantiation of a New Dog

Features of OOP



1. Encapsulation

- Hides internal details of an object
- Protects data from unauthorized access

2. Inheritance

- Creates new classes from existing ones
- Child classes inherit properties and methods from parent classes

J2EE vs. .NET

J2EE

- Based on Java
- Runs on Windows, Linux, and UNIX
- Flexible for large applications
- Good scalability and security

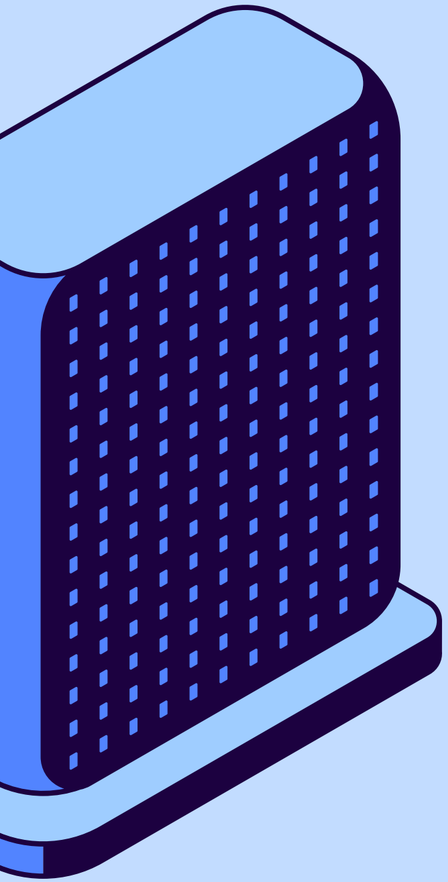
.NET

- Supports multiples languages
- Primarily runs on Windows
- Easier for Microsoft development
- Built-in XML support

Languages for Developing Web Applications

- Uses an **n-tier architecture** (Client → Application Server → Database Server)
- Users interact through web forms
- Data is processed on the **server** and stored in a **database**

Common Web Development Technologies



Client-Side

- HTML
- JavaScript
- AJAX

Server-Side

- PHP
- Java Servlets & JSP
- ASP / ASP.NET
- ColdFusion

Common Uses

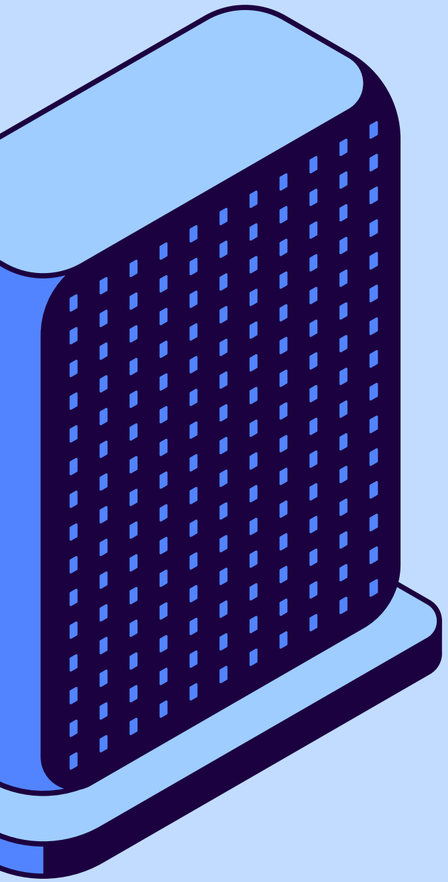
Large Applications

- Java Servlets / JSP
- ASP.NET

Small & Medium Applications

- PHP
- ColdFusion

Why is Object Technology Valuable?



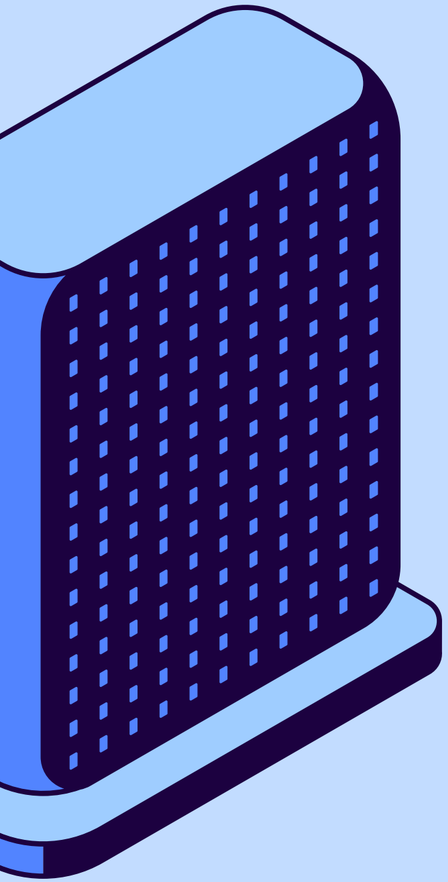
Benefits of Object Technology

- **Productivity –** Faster development through reusable code
- **Maintainability –** Easier to update and fix programs
- **Reusability –** Objects can be used in multiple applications
- **Encapsulation** – Protects data by hiding internal details
- **Inheritance –** Allows new objects to reuse existing features

Database Management System (DBMS)

- Software used to **create**, **manage**, and **protect data**
- Works with the operating system
- Makes data easy to **store**, **retrieve**, and **update**

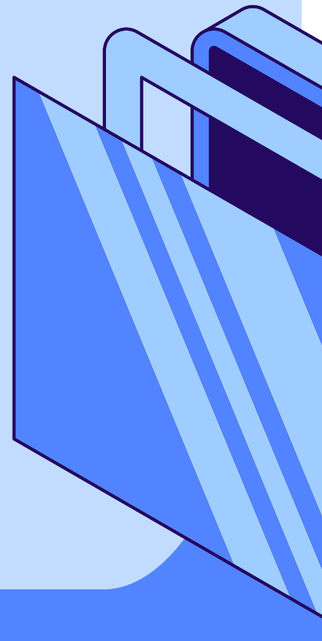
Advantages of DBMS



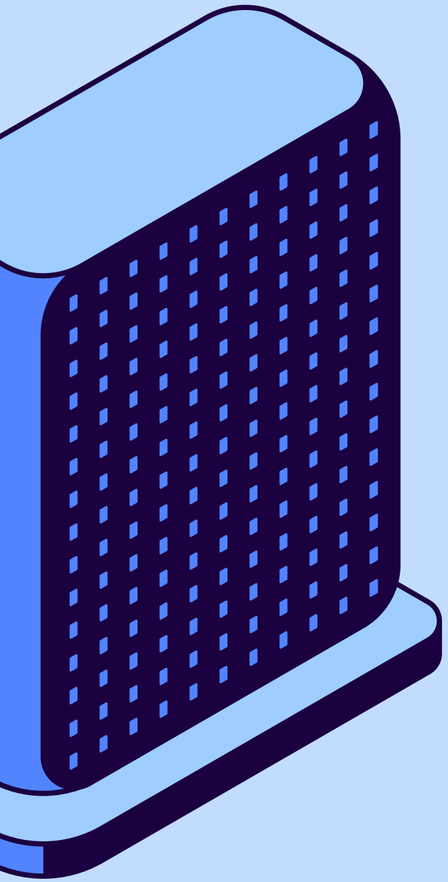
- Easier data retrieval
- Reduces programming errors
- Improves data organization
- Supports **data independence** (changes in storage don’t affect programs)

Types of Database Architectures

Architecture	Description
Hierarchical	Data organized like a tree
Network	Data connected with multiple paths
Relational	Data stored in tables (most common)
Object-Oriented	Stores objects like images, videos, and audio
Object-Relational	Combines relational and object-oriented features



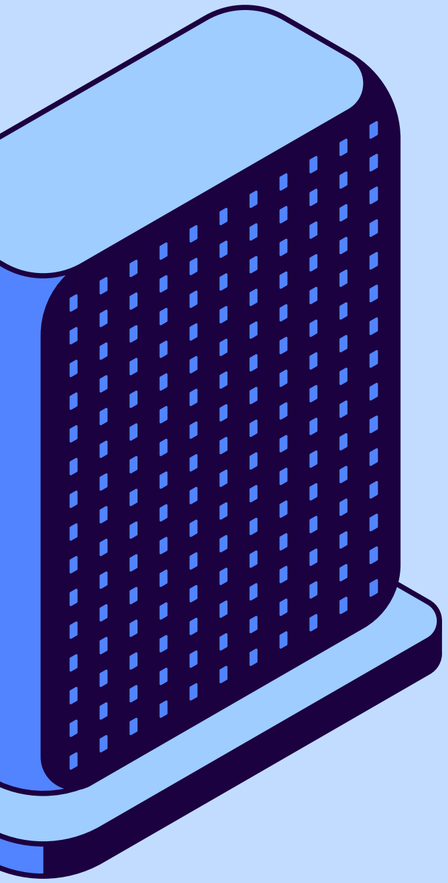
Database Programming



How DBMS is Used

- Uses programming languages like **SQL**
- Allows users to:
  - Search data
  - Insert records
  - Update information
  - Delete records

Other DBMS Tools



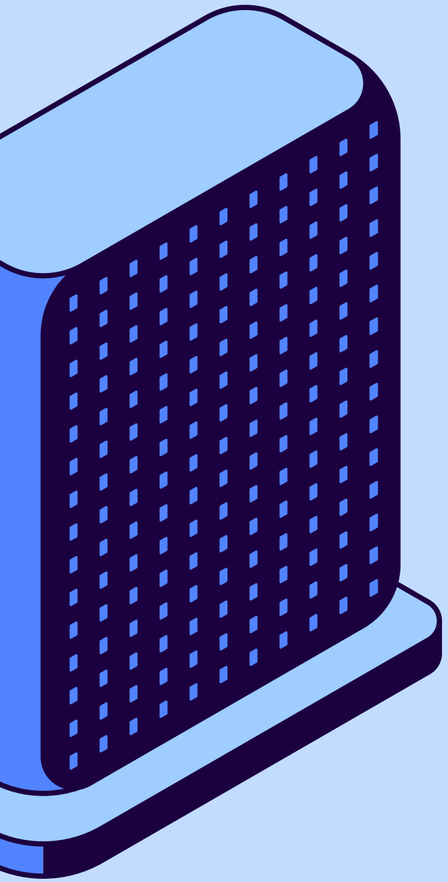
- **Data Dictionary/Directory (DD/D)**
- Stores information about:
  - Data definitions
  - Formats
  - Security
  - Storage location

CASE Tools (Computer-Aided Software Engineering)

What are CASE Tools?

- Software that helps **automate software development**
- Supports different phases of the **Software Development Life Cycle (SDLC**)
- Improves programmer and analyst productivity

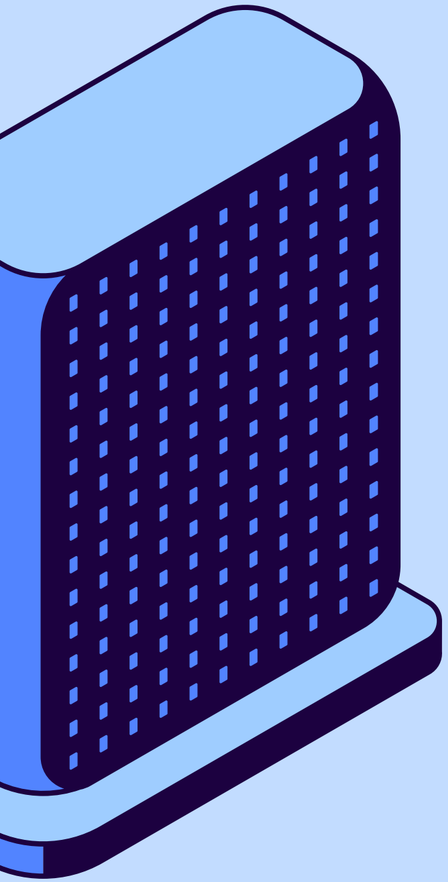
Benefits of CASE Tools



Advantages

- Automates repetitive tasks
- Improves software quality
- Reduces development time
- Helps create diagrams and program code

Modern CASE Tools



UML-Based CASE Tools

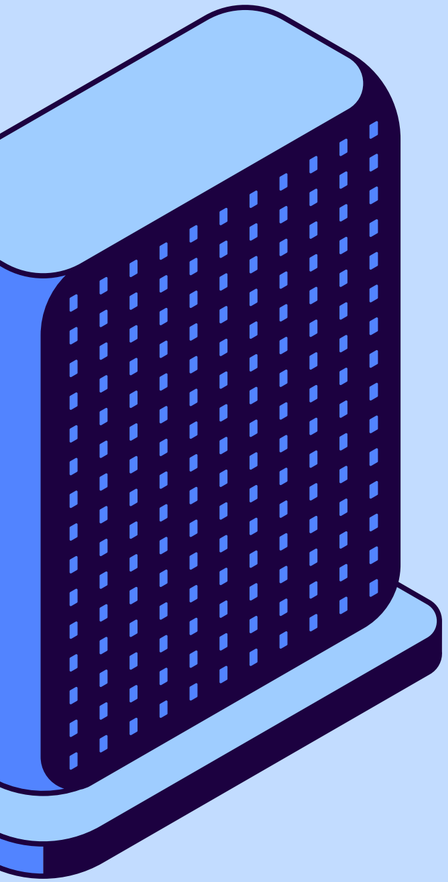
- **Uses Unified Modeling Language (UML)**
- Helps design and visualize software systems
- Examples:
  - IBM Rational Software Architect
  - IBM Rational Rhapsody
  - Borland Together
  - Sybase PowerDesigner

Communications Interface Software

What is it?

- Software that enables **communication between computers**
- Supports **LANs, WANs, and the Internet**
- Allows users to send, receive, and access information

Examples

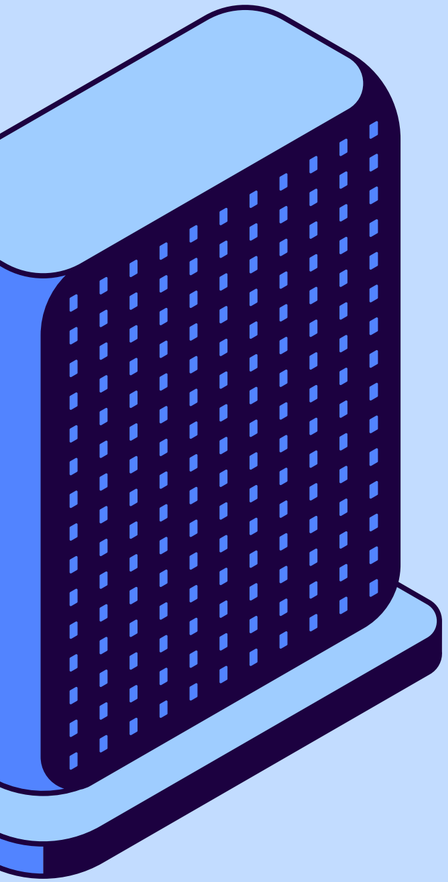


- **Web Browser –** Lets users browse the Internet
- **Web Server Software** – Delivers web pages to users (e.g., Apache)
- **FTP (File Transfer Protocol)** – Transfers files between computers

Utility Programs

- Software that helps **maintain and manage** a computer system
- Performs maintenance and support tasks
- Improves system performance and efficiency

Functions of Utility Programs



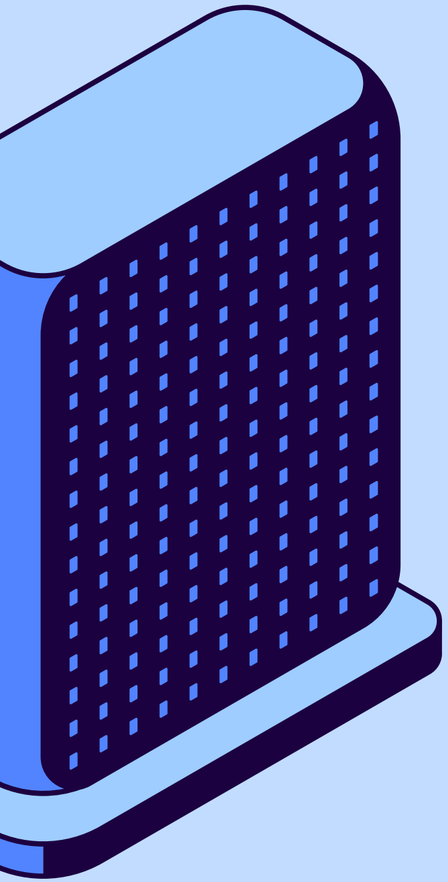
- Load and link programs
- Sort and merge data files
- Copy and manage files
- Compress (zip) and extract (unzip) files
- Detect viruses and spyware
- Organize hard drive storage

The Changing Nature of Software

Key Trends

- Greater focus on **human efficiency**
- More use of **purchased applications**
- Increased use of **Object-Oriented Programming (OOP)**
- More applications running on **Intranets and the Internet**

Other Software Trends



- More **user-developed applications**
- Greater use of **personal productivity software**
  - Web browsers
  - Spreadsheets
  - Database Management Systems (DBMS)

The Information Technology (IT) Industry

What is the IT Industry?

- Includes companies that develop:
  - **Hardware**
  - **Software**
  - **IT Services**
- Two of the largest companies:
  - **Hewlett-Packard (HP)**
  - **IBM**

Major IT Companies

Hardware Companies

- HP
- Dell
- Apple
- Lenovo
- Acer
- Intel (miscroprocessors)

Software Companies

- Microsoft
- Oracle
- SAP
- Adobe
- Symantec

Trends in the IT Industry

Key Trends

- Companies expand through **acquisitions**
- Growth in **software and IT services**
- Strong competition among global technology companies

THANK YOU!

BSAIS 3A

Presented by

Caraga, Danice Izy
Castañeda, Chayris Angel
Timog, Maria Katrina